antibody-binding subunit (resembling the blooms on a bouquet of flowers). C1q is associated with two further subunits, C1r and C1s, in a Ca^{2+}-stabilized trimolecular complex (figure 2.2). Both these molecules contain repeats of a 60-amino acid unit folded as a globular domain and referred to as a complement control protein (CCP) repeat since it is a characteristic structural feature of several proteins involved in control of the complement system. Changes in C1q consequent upon binding the antigen–antibody complex bring about the sequential activation of proteolytic activity in C1r and then C1s.

The next component in the chain, C4 (unfortunately components were numbered before the sequence was established), now binds to C1 through these CCPs and is cleaved enzymically by $\overline{\text{C1s}}$. As expected in a multienzyme cascade, several molecules of C4 undergo cleavage, each releasing a small C4a fragment and revealing a nascent labile internal thiolester bond in the residual $\text{C}\overline{\text{4b}}$ like that in C3 (cf. figure 1.11) which may then bind either to the antibody–C1 complex or the surface of the microbe itself. Note that C4a, like C5a and C3a, has anaphylatoxin activity, although feeble, and C4b resembles C3b in its opsonic activity. In the presence of Mg^{2+}, C2 can complex with the $\text{C}\overline{\text{4b}}$ to become a new substrate for the $\overline{\text{C1s}}$, the resulting product $\overline{\text{C4b2a}}$ now has the vital C3 convertase activity required to cleave C3.

This classical pathway C3 convertase has the same specificity as the $\overline{\text{C3bBb}}$ generated by the alternative pathway, likewise producing the same C3a and C3b fragments. Activation of a single C1 complex can bring about the proteolysis of literally thousands of C3 molecules. From then on things march along exactly in parallel to the post-C3 pathway with one molecule of C3b added to the $\overline{\text{C4b2a}}$ to make it into a C5-splitting enzyme with eventual production of the **membrane attack complex** (figures 1.13 and 2.4). Just as the alternative pathway C3 convertase is controlled by factors H and I, so the breakdown of $\overline{\text{C4b2a}}$ is brought about by either a C4-binding protein (C4bp) or a cell surface C3b receptor (CR1) in the presence of factor I.

The similarities between the two pathways are set out in figure 2.3 and show how antibody can supplement and even improve on the ability of the innate immune system to initiate **acute inflammatory reactions**. Antibody provides yet a further bonus in this respect; the class known as immunoglobulin E (see legend to figure 2.4) can sensitize mast cells through binding to their surface so that combination with antigen triggers mediator release independently of C3a or C5a, adding yet more flexibility to our defenses.

Complexed antibody activates phagocytic cells

We drew attention to the fact that some C3b-coated organisms may adhere to phagocytic cells and yet avoid provoking their uptake. If small amounts of antibody are added the phagocyte springs into action. It does so through the recognition of two or more antibody molecules bound to the microbe, using specialized receptors on the cell surface.

A single antibody molecule complexed to the microorganism is not enough because it cannot cause the cross-linking of antibody receptors in the phagocyte surface membrane which is required to activate the cell. There is a further consideration connected with what is often called **the bonus effect of multivalency**; for thermodynamic reasons, which will be touched on in Chapter 5, the association constant of ligands, which use several rather than a single bond to react with receptors, is increased geometrically rather than arithmetically. For example, three antibodies bound close together on a bacterium could be attracted to a macrophage a thousand times more strongly than a single antibody molecule (figure 2.5).

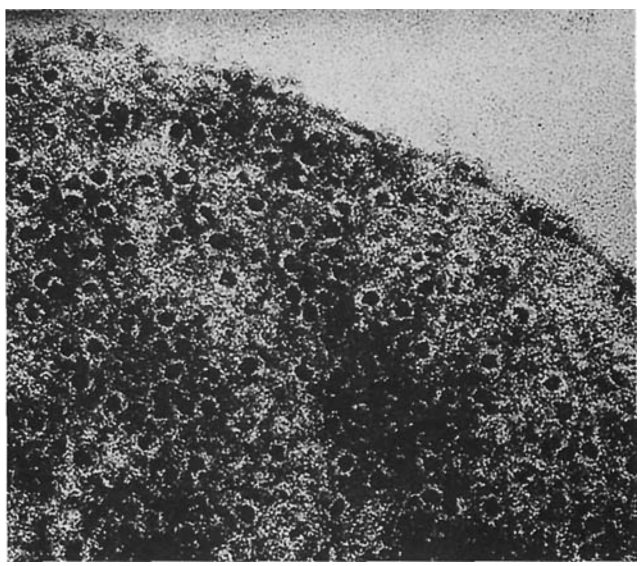


Figure 2.4. Multiple lesions in cell wall of *Escherichia coli* bacterium caused by interaction with IgM antibody and complement. (Human antibodies are divided into five main classes: immunoglobulin M (shortened to IgM), IgG, IgA, IgE and IgD, which differ in the specialization of their 'rear ends' for different biological functions such as complement activation or mast cell sensitization.) Each lesion is caused by a single IgM molecule and shows as a 'dark pit' due to penetration by the 'negative stain'. This is somewhat of an illusion since in reality these 'pits' are like volcano craters standing proud of the surface, and are each single 'membrane attack' complexes. Comparable results may be obtained in the absence of antibody since the cell wall endotoxin can activate the alternative pathway in the presence of higher concentration of serum (×400 000). (Courtesy of Drs R. Dourmashkin and J.H. Humphrey.)

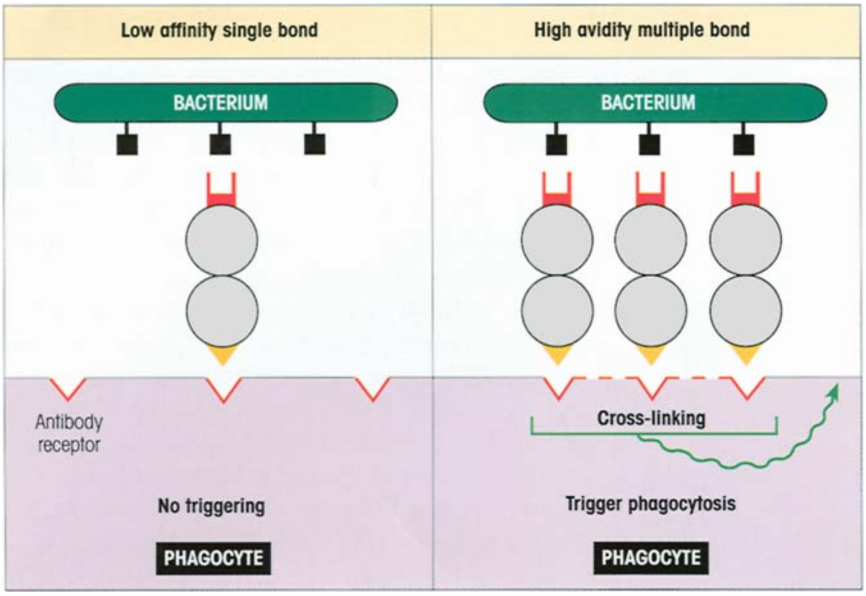


Figure 2.5. Binding of bacterium to phagocyte by multiple antibodies gives strong association forces and triggers phagocytosis by cross-linking the surface receptors for antibody.

CELLULAR BASIS OF ANTIBODY PRODUCTION

Antibodies are made by lymphocytes

The majority of resting **lymphocytes** are small cells with a darkly staining nucleus due to condensed chromatin and relatively little cytoplasm containing the odd mitochondrion required for basic energy provision. Figures 2.6 and 2.7 compare the morphology of these cells with that of the minority population of **large granular lymphocytes** which includes the natural killer (NK) set referred to in Chapter 1.

The central role of the **small lymphocyte** in the production of antibody was established largely by the work of Gowans. He depleted rats of their lymphocytes by chronic drainage of lymph from the thoracic duct by an indwelling cannula, and showed that they had a grossly impaired ability to mount an antibody response to microbial challenge. The ability to form antibody could be restored by injecting thoracic duct lymphocytes obtained from another rat. The same effect could be obtained if, before injection, the thoracic duct cells were first incubated at 37°C for 24 hours under conditions which kill off large- and medium-sized cells and leave only the small lymphocytes. This shows that the small lymphocyte is necessary for the **antibody response**.

The small lymphocytes can be labeled if the donor rat is previously injected with tritiated thymidine; it then becomes possible to follow the fate of these lymphocytes when transferred to another rat of the same strain which is then injected with microorganisms to produce an antibody response (figure 2.8). It transpires that after contact with the injected microbes, some of the transferred labeled lymphocytes develop into **plasma cells** (figures 2.6d and 2.9) which can be shown to contain (figure 2.6e) and secrete antibody.

Antigen selects the lymphocytes which make antibody

The molecules in the microorganisms which evoke and react with antibodies are called **antigens** (**gen**erates **anti**bodies). We now know that antibodies are formed before antigen is ever seen and that they are **selected** for by antigen.

It works in the following way. Each lymphocyte of a subset called the **B-lymphocytes**, because they differentiate in the *bone marrow*, is programed to make one, and only one, antibody and it places this antibody on its outer surface to act as a receptor. This can be detected by using fluorescent probes and, in figure 2.6c, one can see the molecules of antibody on the surface of a human B-lymphocyte stained with a fluorescent rabbit antiserum raised against a preparation of human antibodies. Each lymphocyte has of the order of 10^5 identical antibody molecules on its surface.

When an antigen enters the body, it is confronted by a dazzling array of lymphocytes all bearing different antibodies each with its own individual recognition site. The antigen will only bind to those receptors with which it makes a good fit. Lymphocytes whose recep-

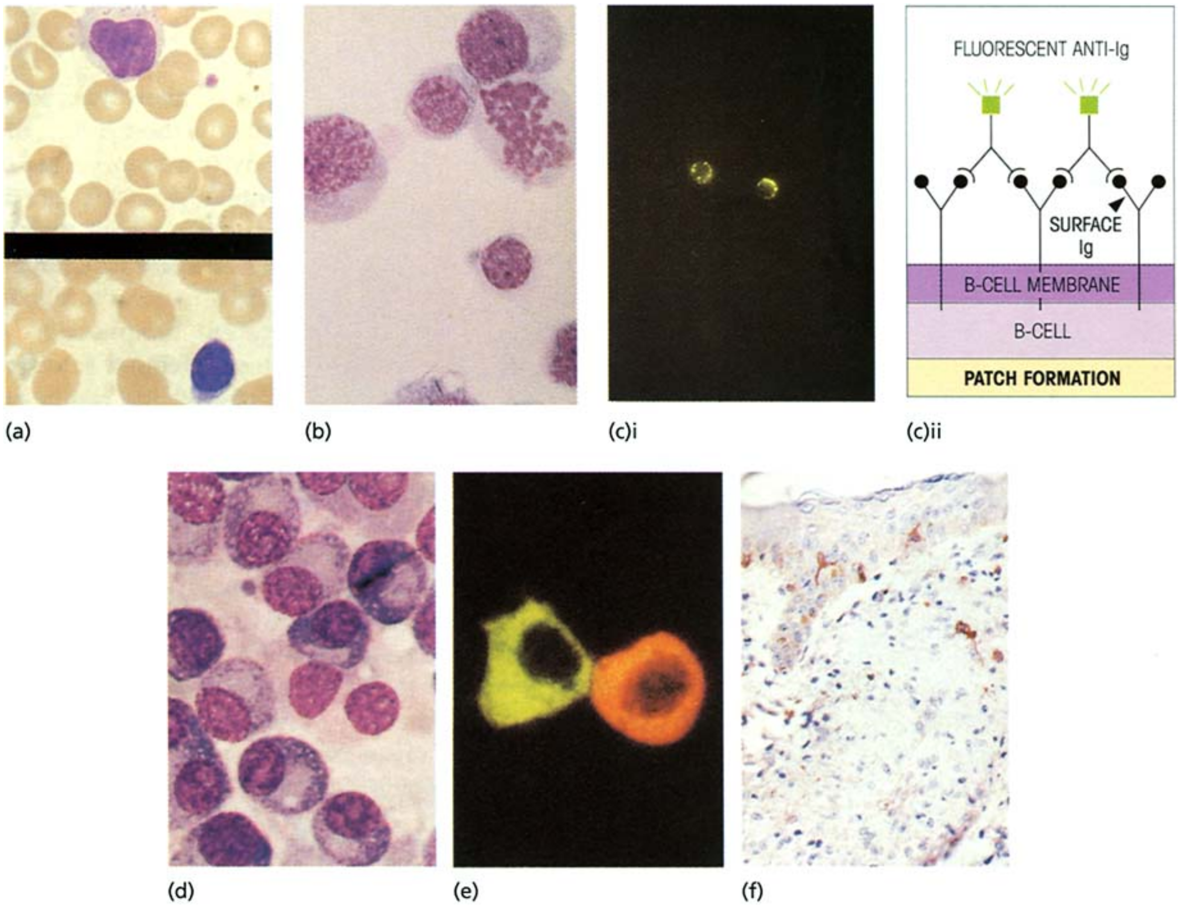
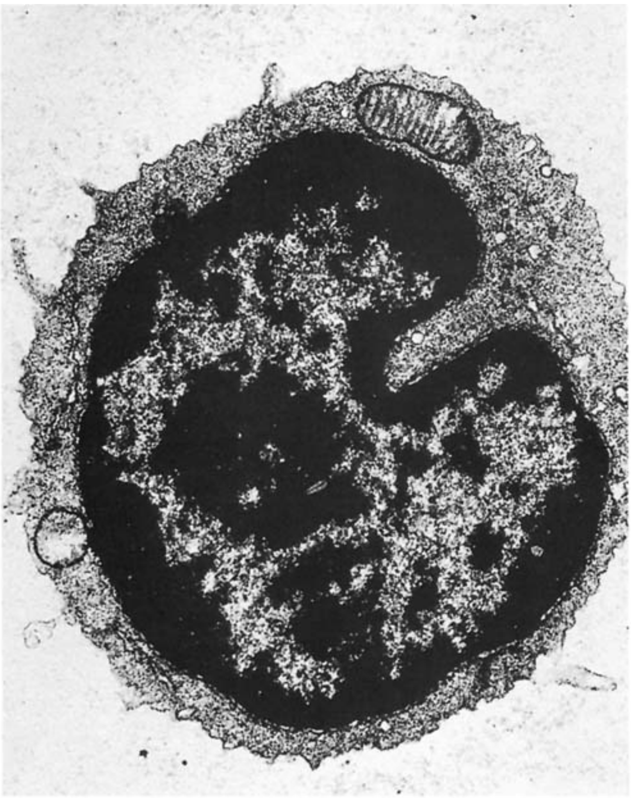


Figure 2.6. Cells involved in the acquired immune response. (a) Small lymphocytes. Condensed chromatin gives rise to heavy staining of the nucleus. The cell on the bottom is a typical resting agranular T-cell with a thin rim of cytoplasm. The upper nucleated cell is a large granular lymphocyte; it has more cytoplasm and azurophilic granules are evident. Isolated platelets are visible. B-lymphocytes range from small to intermediate in size and lack granules. Giemsa stain. (b) Transformed T-lymphocytes (lymphoblasts) following stimulation of lymphocytes in culture with a polyclonal activator, such as the lectins phytohemagglutinin, concanavalin A and pokeweed mitogen which stimulate a wide range of cells independently of their specificity for antigen. The large lymphoblasts with their relatively high ratio of cytoplasm to nucleus may be compared in size with the isolated small lymphocyte. One cell is in mitosis. May–Grünwald–Giemsa. (c) Immunofluorescent staining of B-lymphocyte surface immunoglobulin using fluorescein-conjugated (■) anti-Ig. Provided the reaction is carried out in the cold to prevent pinocytosis, the labeled antibody cannot penetrate to the interior of the viable lymphocytes and reacts only with surface components. Patches of aggregated surface Ig are seen which are beginning to form a cap in the right-hand lymphocyte. During cap formation, submembranous myosin becomes redistributed in association with the surface Ig and induces locomotion of the previously sessile cell in a direction away from the cap. (d) Plasma cells. The nucleus is eccentric. The cytoplasm is strongly basophilic due to high RNA content. The juxtanuclear lightly stained zone corresponds with the Golgi region. May–Grünwald–Giemsa. (e) Plasma cells stained to show intracellular immunoglobulin using a fluorescein-labeled anti-IgG (green) and a rhodamine-conjugated anti-IgM (red). (f) Langerhans' cells in human epidermis in leprosy, increased in the subepidermal zone, possibly as a consequence of the disease process. Stained by the immunoperoxidase method with S-100 antibodies. (Material for (a) was kindly supplied by Mr M. Watts of the Department of Haematology, Middlesex Hospital Medical School; (b) and (c) by Professor P. Lydyard; (d) and (e) by Professor C. Grossi; and (f) by Dr Marian Ridley.)

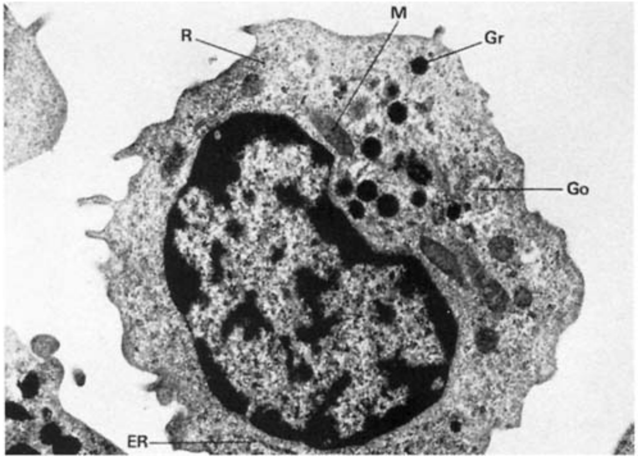
tors have bound antigen receive a triggering signal and develop into antibody-forming plasma cells and, since the lymphocytes are programed to make only one antibody, that secreted by the plasma cell will be identical with that originally acting as the lymphocyte receptor, i.e. it will bind well to the antigen. In this way, antigen selects for the antibodies which recognize it effectively (figure 2.10).

The need for clonal expansion means humoral immunity must be acquired

Because we can make hundreds of thousands, maybe even millions, of different antibody molecules, it is not feasible for us to have too many lymphocytes producing each type of antibody; there just would not be enough room in the body to accommodate them. To



(a)



(b)

Figure 2.7. Lymphocyte ultrastructure. (a) Small agranular T-lymphocyte. Indented nucleus with condensed chromatin, sparse cytoplasm: single mitochondrion shown and many free ribosomes but otherwise few organelles (×13 000). B-lymphocytes are essentially similar with slightly more cytoplasm and occasional elements of rough-surfaced endoplasmic reticulum. (b) Large granular lymphocyte (×7500). The more abundant cytoplasm contains several mitochondria (M), free ribosomes (R) with some minor elements of rough-surfaced endoplasmic reticulum (ER), prominent Golgi apparatus (Go) and characteristic membrane-bound electron-dense granules (Gr). The nuclear chromatin is less condensed than that of the agranular T-cell. (Courtesy of Professors A. Zicca and C.E. Grossi.)

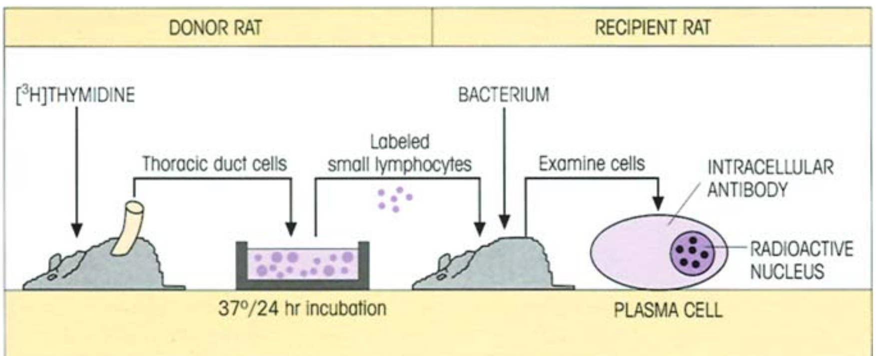


Figure 2.8. Labeled small lymphocytes become antibody-forming plasma cells when transferred to a recipient rat which is immunized with a bacterium. Transferred cell with radioactive nucleus shown by autoradiography. Intracellular antibody revealed by staining with fluorescent probes (cf. figure 2.6e).

compensate for this, lymphocytes which are triggered by contact with antigen undergo successive waves of proliferation (figure 2.6b) to build up a large clone of plasma cells which will be making antibody of the kind for which the parent lymphocyte was programed. By this system of **clonal selection**, large enough concentrations of antibody can be produced to combat infection effectively (Milestone 2.1; figure 2.11).

The importance of proliferation for the development of a significant antibody response is highlighted by the ability of antimitotic drugs to abolish antibody production to a given antigen stimulus completely.

Because it takes time for the proliferating clone to build up its numbers sufficiently, it is usually several days before antibodies are detectable in the serum following primary contact with antigen. The newly formed antibodies are a consequence of antigen exposure and it is for this reason that we speak of the **acquired immune response**.

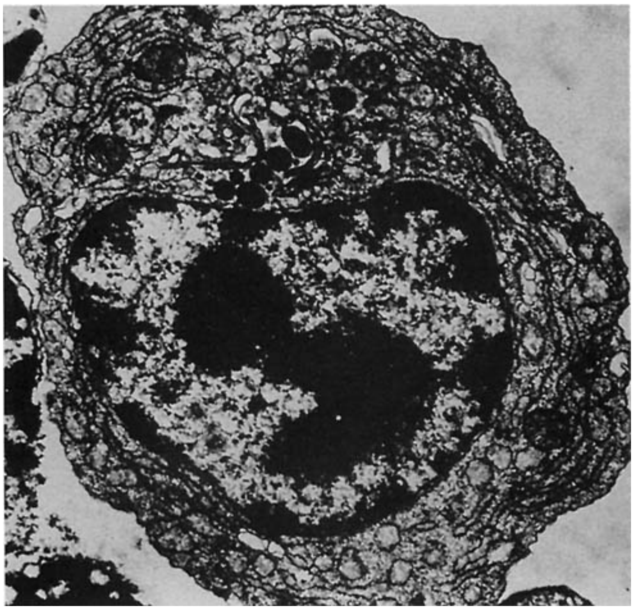


Figure 2.9. Plasma cell (×10 000). Prominent rough-surfaced endoplasmic reticulum associated with the synthesis and secretion of Ig.

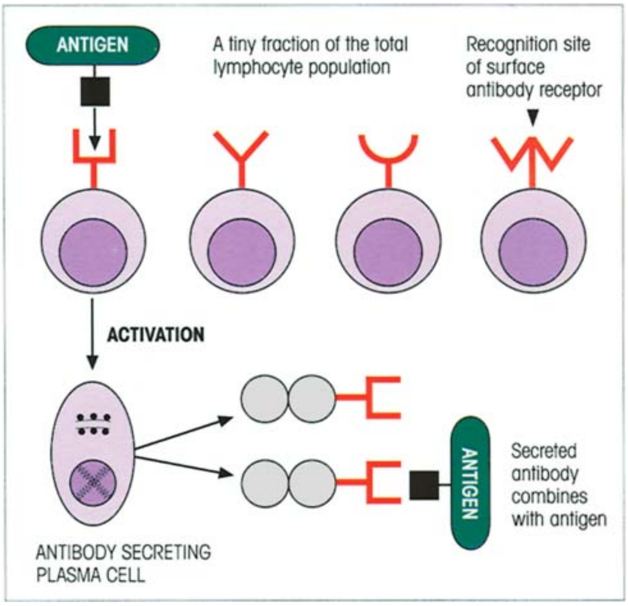


Figure 2.10. Antigen activates those B-cells whose surface antibody receptors it can combine with firmly.

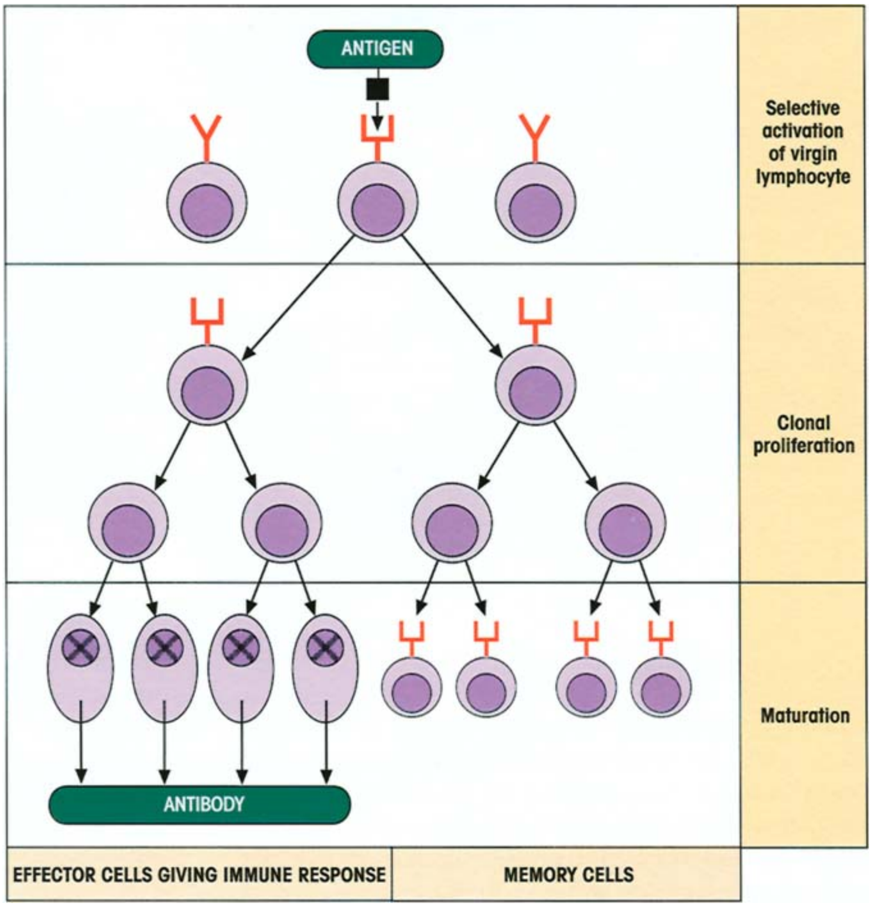


Figure 2.11. The cellular basis for the generation of effector and memory cells by clonal selection after primary contact with antigen. The cell selected by antigen undergoes many divisions during the clonal proliferation and the progeny mature to give an expanded population of antibody-forming cells. The antibody response is particularly vulnerable to antimitotic agents at the proliferation stage. A fraction of the progeny of the original antigen-reactive lymphocytes become nondividing memory cells and others the effector cells of either humoral, i.e. antibody-mediated, or, as we shall see subsequently, cell-mediated immunity. Memory cells require fewer cycles before they develop into effectors and this shortens the reaction time for the secondary response. The expanded clone of cells with memory for the original antigen provides the basis for the greater secondary relative to the primary immune response. Priming with low doses of antigen can often stimulate effective memory without producing very adequate antibody synthesis.

Milestone 2.1—Clonal Selection Theory

Antibody production according to Ehrlich

In 1894, well in advance of his time as usual, the remarkable Paul Ehrlich proposed the side-chain theory of antibody production. Each cell would make a large variety of surface receptors which bound foreign antigens by complementary shape 'lock and key' fit. Exposure to antigen would provoke overproduction of receptors (antibodies) which would then be shed into the circulation (figure M2.1.1).

Template theories

Ehrlich's hypothesis implied that antibodies were preformed prior to antigen exposure. However, this view was difficult to accept when later work showed that antibodies could be formed to almost any organic structure synthesized in the chemist's laboratory (e.g. azobenzene arsonate; figure 5.1) despite the fact that such molecules would never be encountered in the natural environment. Thus was born the idea that antibodies were synthesized by using the antigen as a template. Twenty years passed before this idea was 'blown out of the water' by the observation that, after an antibody molecule is unfolded by guanidinium salts in the absence of antigen, it spontaneously refolds to regenerate its original specificity. It became clear that each antibody has a different amino acid sequence which governs its final folded shape and hence its ability to recognize antigen.

Selection theories

The wheel turns full circle and we once more live with the idea that, since different antibodies must be encoded by separate genes, the information for making these antibodies must pre-exist in the host DNA. In 1955, Nils Jerne perceived that this could form the basis for a selective theory of antibody production. He suggested that the complete antibody repertoire is expressed at a low level and that, when antigen enters the body, it selects its complementary antibody to form a complex which in some way provokes further synthesis of that particular antibody. But how?

Mac Burnet now brilliantly conceived of a cellular basis for this selection process. Let each lymphocyte be programed to make its own singular antibody which is inserted like an Ehrlich 'side-chain' into its surface membrane. Antigen will now form the complex envisaged by Jerne, on the surface of the lymphocyte, and by triggering its activation and clonal proliferation, large amounts of the specific antibody will be synthesized (figure 2.11). Bow graciously to that soothsayer Ehrlich—he came so close in 1894!

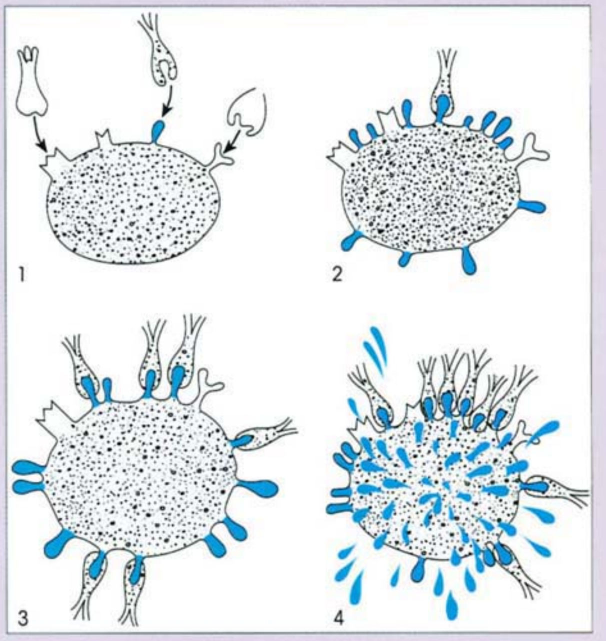


Figure M2.1.1. Ehrlich's side-chain theory of Ab production. (Reproduced from *Proceedings of the Royal Society B* (1900), **66**, 424.)

ACQUIRED MEMORY

When we make an antibody response to a given infectious agent, by definition that microorganism must exist in our environment and we are likely to meet it again. It would make sense then for the immune mechanisms alerted by the first contact with antigen to leave behind some memory system which would enable the response to any subsequent exposure to be faster and greater in magnitude.

Our experience of many common infections tells us that this must be so. We rarely suffer twice from such diseases as measles, mumps, chickenpox, whooping cough and so forth. The first contact clearly imprints some information, imparts some **memory**, so that the body is effectively prepared to repel any later invasion by that organism and a state of immunity is established.

Secondary antibody responses are better

By following the production of antibody on the first and second contacts with antigen, we can see the basis for the development of immunity. For example, when we inject a bacterial product such as tetanus toxoid into a rabbit, for the reasons already discussed, several days elapse before antibodies can be detected in the blood; these reach a peak and then fall (figure 2.12). If we now allow the animal to rest and then give a second injection of toxoid, the course of events is dramatically altered. Within 2–3 days the antibody level in the blood rises steeply to reach much higher values than were observed in the **primary response**. This **secondary response** then is characterized by a more rapid and more abundant production of antibody resulting from the 'tuning up' or priming of the antibody-forming system.

With our knowledge of lymphocyte function, it is perhaps not surprising to realize that these are the cells which provide memory. This can be demonstrated by **adoptive transfer** of lymphocytes to another animal, an experimental system frequently employed in immunology (cf. figure 2.8). In the present case, the immunologic potential of the transferred cells is expressed in a recipient treated with X-rays which destroy its own lymphocyte population; thus any immune response will be of donor not recipient origin. In the experiment described in figure 2.13, small lymphocytes are taken from an animal given a primary injection of tetanus toxoid and transferred to an irradiated host which is then boosted with the antigen; a rapid, intense production of antibody characteristic of a secondary response is seen. To exclude the possibility that the first antigen injection might exert a *nonspecific* stimulatory effect on the lymphocytes, the boosting injection includes influenza hemagglutinin as a control antigen. Furthermore, a 'criss-cross' control group primed with influenza hemagglutinin must also be included to ensure that this antigen is capable of giving a secondary booster response. We have explained the design of the experiment at some length to call attention to the need for careful selection of controls.

The higher response given by a primed lymphocyte population can be ascribed mainly to an expansion of the numbers of cells capable of being stimulated by the antigen (figure 2.11), although we shall see later that there are some qualitative differences in these memory cells as well (pp. 195–197).

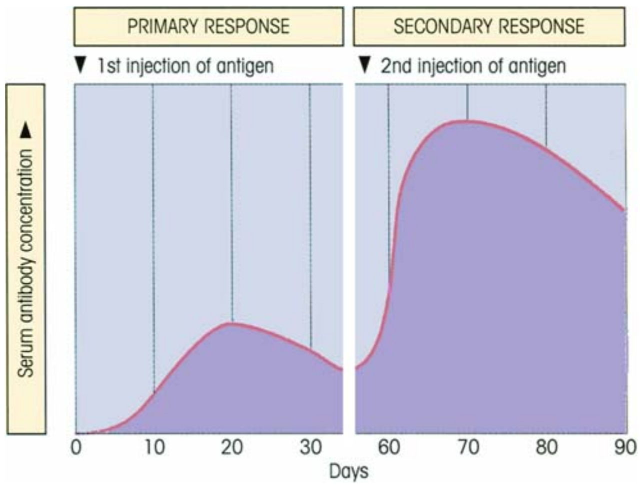


Figure 2.12. Primary and secondary response. A rabbit is injected on two separate occasions with tetanus toxoid. The antibody response on the second contact with antigen is more rapid and more intense.

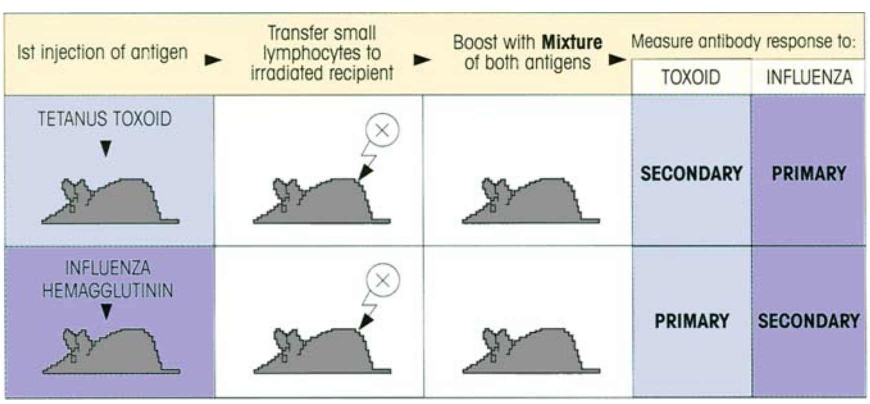


Figure 2.13. Memory for a primary response can be transferred by small lymphocytes. Recipients are treated with a dose of X-rays which directly kill lymphocytes (highly sensitive to radiation) but only affect other body cells when they divide; the recipient thus functions as a living 'test-tube' which permits the function of the donor cells to be followed. The reasons for the design of the experiment are given in the text. In practice, because of the possibility of interference between the two antigens, it would be wiser to split each of the primary antigen-injected groups into two, giving a separate boosting antigen to each to avoid using a mixture.

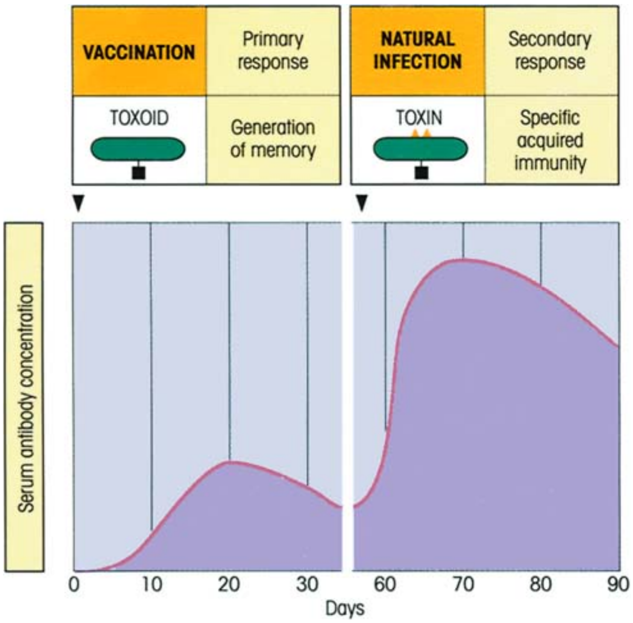


Figure 2.14. The basis of vaccination illustrated by the response to tetanus toxoid. Treatment of the bacterial toxin with formaldehyde destroys its toxicity (associated with ▲▲) but retains antigenicity. Exposure to toxin in a subsequent natural infection boosts the memory cells, producing high levels of neutralizing antibody which are protective.

ACQUIRED IMMUNITY HAS ANTIGEN SPECIFICITY

Discrimination between different antigens

The establishment of memory or immunity by one organism does not confer protection against another unrelated organism. After an attack of measles we are immune to further infection but are susceptible to other agents such as the polio or mumps viruses. Acquired immunity then shows **specificity** and the immune system can differentiate specifically between the two organisms. A more formal experimental demonstration of this discriminatory power was seen in figure 2.13 where priming with tetanus toxoid evoked memory for that antigen but not for influenza and vice versa.

The basis for this lies of course in the ability of the recognition sites of the antibody molecules to distinguish between antigens; antibodies which react with the toxoid do not bind to influenza and, *mutatis mutandis* as they say, anti-influenza is not particularly smitten with the toxoid.

Discrimination between self and nonself

This ability to recognize one antigen and distinguish it from another goes even further. The individual must also recognize what is foreign, i.e. what is 'nonself'. The failure to discriminate between **self** and **nonself** could lead to the synthesis of antibodies directed against components of the subject's own body (**autoantibodies**), which in principle could prove to be highly embarrassing. On purely theoretical grounds it seemed to Burnet and Fenner that the body must develop some mechanism whereby 'self' and 'nonself' could be distinguished, and they postulated that those circulating body components which were able to reach the developing lymphoid system in the perinatal period could in some way be 'learnt' as 'self'. A permanent unresponsiveness or **tolerance** would then be created so that as immunologic maturity was reached there would normally be an inability to respond to 'self' components. At this stage it is salutory to note that Burnet had the sagacity to realize that his clonal selection theory could readily provide the cellular basis for such a mechanism to operate. He argued that if each lymphocyte were preoccupied with making its own individual antibody, those cells programed to express antibodies reacting with circulating self components could be rendered unresponsive without affecting those lymphocytes specific for foreign antigens. In other words, self-reacting lymphocytes could be selectively suppressed or tolerized without undermining the ability of the host to respond immunologically to infectious agents. As we shall see in Chapter 12, these predictions have been amply verified, although we will learn that, as new lymphocytes differentiate throughout life, they will all go through this self-tolerizing screening process. However, self tolerance is not absolute and normally innocuous but potentially harmful anti-self lymphocytes exist in all of us.

VACCINATION DEPENDS ON ACQUIRED MEMORY

Some 200 years ago, Edward Jenner carried out the remarkable studies which mark the beginning of immunology as a systematic subject. Noting the pretty pox-free skin of the milkmaids, he reasoned that deliberate exposure to the pox virus of the cow, which is not virulent for the human, might confer protection against the related human smallpox organism. Accordingly, he inoculated a small boy with cowpox and was delighted—and presumably breathed a sigh of relief—to observe that the boy was now protected against a subsequent exposure to smallpox (what would today's ethical committees have said about that?!). By injecting a harmless form of a disease organism, Jenner had utilized the specificity and memory of the acquired immune response to lay the foundations for modern **vaccination** (Latin *vacca*, cow).

The essential strategy is to prepare an *innocuous* form of the infectious organism or its toxins which still substantially retains the antigens responsible for establishing protective immunity. This has been done by using killed or live attenuated organisms, purified microbial components or chemically modified antigens (figure 2.14).

CELL-MEDIATED IMMUNITY PROTECTS AGAINST INTRACELLULAR ORGANISMS

Many microorganisms live inside host cells where it is impossible for humoral antibody to reach them. Obligate intracellular parasites like viruses have to replicate inside cells; facultative intracellular parasites like *Mycobacteria* and *Leishmania* can replicate within cells, particularly macrophages, but do not have to; they like the intracellular life because of the protection it affords. A totally separate acquired immunity system has evolved to deal with this situation based on a distinct lymphocyte subpopulation made up of **T-cells**, designated thus because, unlike the B-lymphocytes, they differentiate within the milieu of the **thymus gland**. Because they are specialized to operate against cells bearing intracellular organisms, T-cells only recognize antigen when it is on the surface of a body cell. Accordingly, the **T-cell surface receptors**, which are different from the antibody molecules used by B-lymphocytes, recognize antigen plus a surface marker which informs the T-lymphocyte that it is making contact with another cell. These cell markers belong to an important group of molecules known as the **major histocompatibility complex (MHC)**, identified originally through their ability to evoke powerful transplantation reactions in other members of the same species. Now naive or virgin T-cells must be introduced to the antigen and MHC by a special dendritic antigen-presenting cell (figures 2.6f and 8.13) before they can be initiated into the rites of a primary response. However, once primed, they are activated by antigen and MHC present on the surface of other cell types such as macrophages as we shall now see.

Cytokine-producing T-cells help macrophages to kill intracellular parasites

These organisms only survive inside macrophages through their ability to subvert the innate killing mechanisms of these cells. Nonetheless, they cannot prevent the macrophage from processing small antigenic fragments (possibly of organisms which have spontaneously died) and placing them on the host cell surface. A subpopulation of T-lymphocytes called **T-helper cells**, if primed to that antigen, will recognize and bind to the combination of antigen with so-called **class II** MHC molecules on the macrophage surface and produce a variety of soluble factors termed **cytokines** which include the interleukins IL-2, etc. (p. 177). Different cytokines can be made by various cell types and generally act at a short range on neighboring cells. Some T-cell cytokines help B-cells to make antibodies, while others such as γ-interferon (IFNγ) act as **macrophage activating factors** which switch on the previously subverted microbicidal mechanisms of the macrophage and bring about the death of the intracellular microorganisms (figure 2.15).

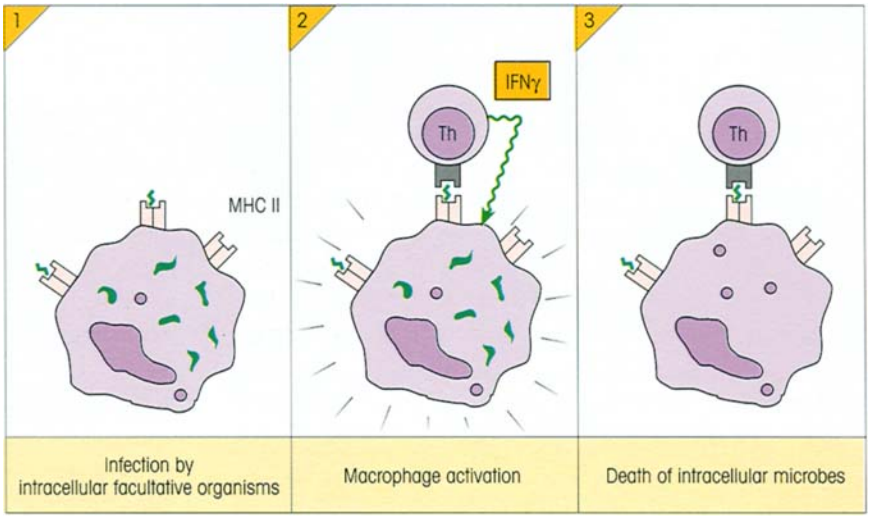


Figure 2.15. Intracellular killing of microorganisms by macrophages. (1) Surface antigen (⌇) derived from the intracellular microbes is complexed with class II MHC molecules ( ). (2) The primed T-helper cell binds to this surface complex and is triggered to release the cytokine γ-interferon (IFNγ). This activates microbicidal mechanisms in the macrophage. (3) The infectious agent meets a timely death.

Virally infected cells can be killed by cytotoxic T-cells and ADCC

We have already discussed the advantage to the host of killing virally infected cells before the virus begins to replicate and have seen that large granular lymphocytes with NK activity (p. 18) can subserve a cytotoxic function. However, NK cells have a limited range of specificities and, in order to improve their efficacy, this range needs to be expanded.

One way in which this can be achieved is by coating the target cell with antibodies specific for the virally coded surface antigens because NK cells have receptors for the constant part of the antibody molecule, rather like phagocytic cells. Thus antibodies will bring the NK cell very close to the target by forming a bridge, and the NK cell being activated by the complexed antibody molecules is able to kill the virally infected cell by its extracellular mechanisms (figure 2.16). This system, termed **antibody-dependent cellular cytotoxicity (ADCC)**, is very impressive when studied *in vitro* but it has proved difficult to establish to what extent it operates within the body.

On the other hand, a **subset** of **T-cells with cytotoxic potential** has evolved for which there is clear evidence of *in vivo* activity. Like the T-helpers, these cells have a very wide range of antigen specificities because they clonally express a large number of different surface receptors similar to, but not identical with, the surface antibody receptors on the B-lymphocytes. Again, each lymphocyte is programed to make only one receptor and, again like the T-helper cell, recognizes antigen only in association with a cell marker, in this case the **class I** MHC molecule (figure 2.16). Through this recognition of surface antigen, the cytotoxic cell comes into intimate contact with its target and administers the 'kiss of apoptotic death'. It also releases **IFNγ** which would help to reduce the spread of virus to adjacent cells, particularly in cases where the virus itself may prove to be a weak inducer of IFNα or β.

In an entirely analogous fashion to the B-cell, T-cells are selected and activated by combination with antigen, expanded by clonal proliferation and mature to give T-helpers and cytotoxic T-effectors, together with an enlarged population of memory cells. Thus both T- and B-cells provide **specific acquired immunity** with a variety of mechanisms, which in most cases operate to extend the range of effectiveness of innate immunity and confer the valuable advantage that a first infection

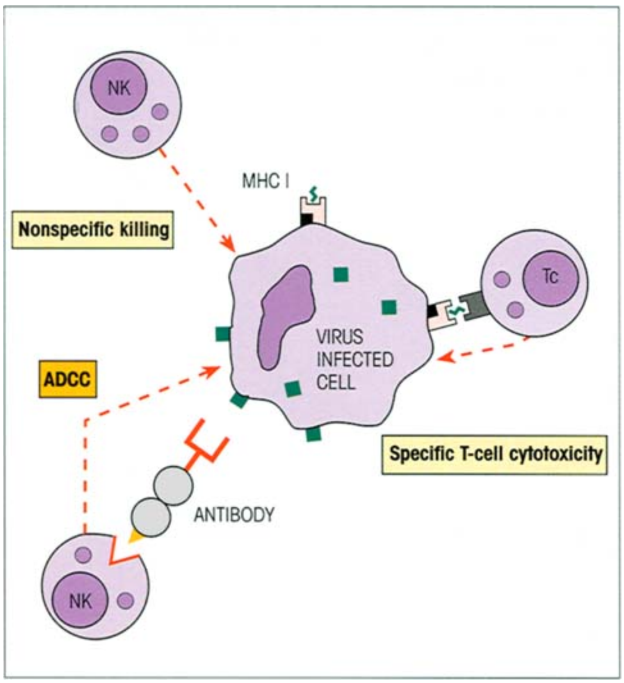


Figure 2.16. Killing virally infected cells. The nonspecific killing mechanism of the NK cell can be focused on the target by antibody to produce antibody-dependent cellular cytotoxicity (ADCC). The cytotoxic T-cell homes onto its target specifically through receptor recognition of surface antigen in association with MHC class I molecules.

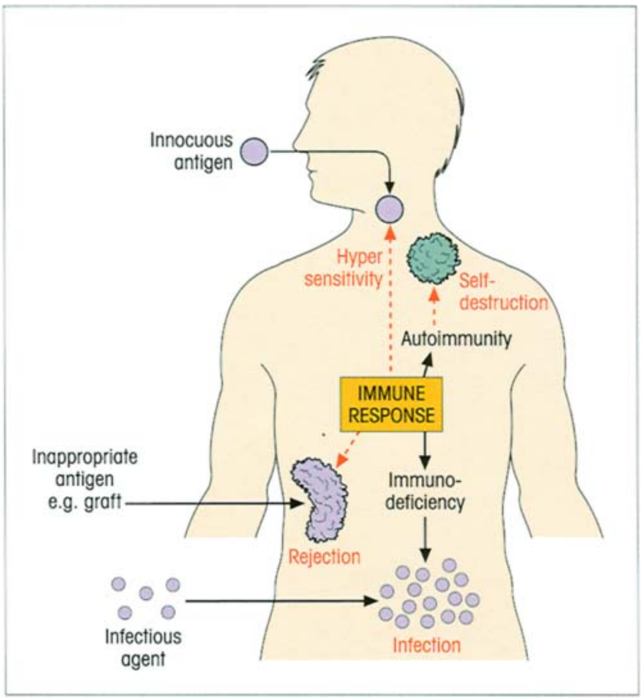


Figure 2.17. Inappropriate immune responses can produce damaging reactions such as the hypersensitivity response to inhaled otherwise innocuous allergens, the destruction of self tissue by autoimmune attack, the rejection of tissue transplants, and the susceptibility to infection in immunodeficient individuals.

prepares us to withstand further contact with the same microorganism.

IMMUNOPATHOLOGY

The immune system is clearly 'a good thing', but like mercenary armies, it can turn to bite the hand that feeds it, and cause damage to the host (figure 2.17).

Thus where there is an especially heightened response or persistent exposure to exogenous antigens, tissue damaging or **hypersensitivity** reactions may result. Examples are allergy to grass pollens, blood dyscrasias associated with certain drugs, immune complex glomerulonephritis occurring after streptococcal infection, and chronic granulomas produced during tuberculosis or schistosomiasis.

In other cases, hypersensitivity to autoantigens may arise through a breakdown in the mechanisms which control self tolerance, and a wide variety of **autoimmune diseases**, such as insulin-dependent diabetes and multiple sclerosis and many of the rheumatologic disorders, have now been recognized.

Another immunopathologic reaction of some consequence is **transplant rejection**, where the MHC antigens on the donor graft may well provoke a fierce reaction. Lastly, one should consider the by no means infrequent occurrence of inadequate functioning of the immune system—**immunodeficiency**. We would like to think that at this stage the reader would have no difficulty in predicting that the major problems in this condition relate to persistent infection, the type of infection being related to the elements of the immune system which are defective.

SUMMARY

Antibody—the specific adaptor

- The antibody molecule evolved as a specific adaptor to attach to microorganisms which either fail to activate the alternative complement pathway or prevent activation of the phagocytic cells.
- The antibody fixes to the antigen by its specific recognition site and its constant structure regions activate complement through the classical pathway (binding C1 and generating a $\overline{C4b2a}$ convertase to split C3) and phagocytes through their antibody receptors.
- This supplementary route into the acute inflammatory reaction is enhanced by antibodies which sensitize mast cells and by immune complexes which stimulate mediator release from tissue macrophages (figure 2.18).

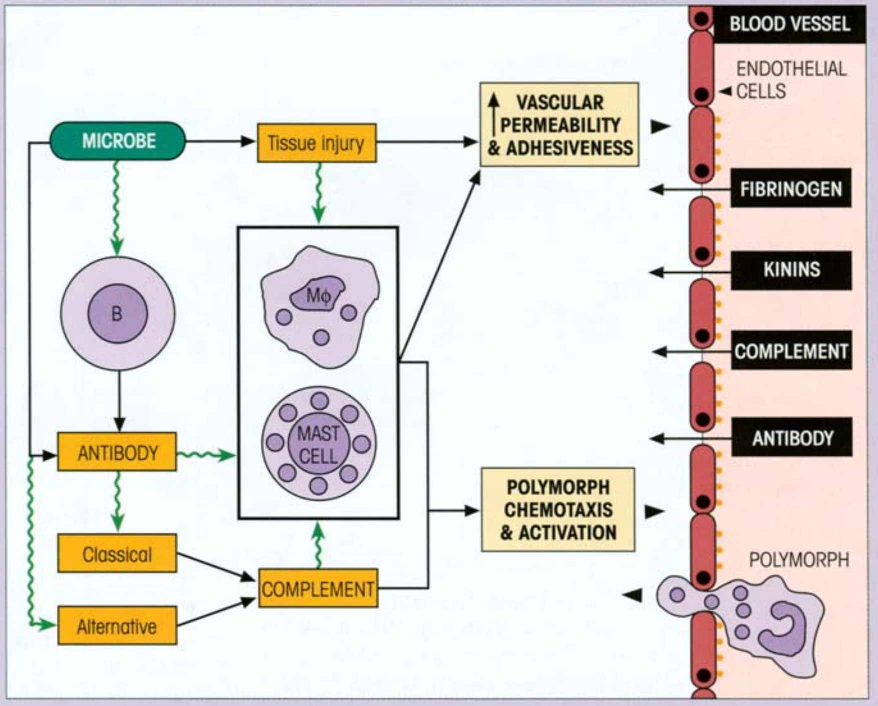


Figure 2.18. Production of a protective acute inflammatory reaction by microbes either: (i) through tissue injury (e.g. bacterial toxin) or direct activation of the alternative complement pathway, or (ii) by antibody-dependent triggering of the classical complement pathway or mast cell degranulation (a special class of antibody, IgE, does this).

Cellular basis of antibody production

• Antibodies are made by plasma cells derived from B-lymphocytes, each of which is programed to make antibody of a single specificity which is placed on the cell surface as a receptor.

• Antigen binds to the cell with a complementary antibody, activates it and causes clonal proliferation and finally maturation to antibody-forming cells and memory cells. Thus the antigen brings about clonal selection of the cells making antibody to itself.

Acquired memory and vaccination

• The increase in memory cells after priming means that the acquired secondary response is faster and greater, providing the basis for vaccination using a harmless form of the infective agent for the initial injection.

Acquired immunity has antigen specificity

• Antibodies differentiate between antigens because recognition is based on molecular shape complementarity. Thus memory induced by one antigen will not extend to another unrelated antigen.

• The immune system differentiates self components from foreign antigens by making immature self-reacting lymphocytes unresponsive through contact with host molecules; lymphocytes reacting with foreign antigens are unaffected since they only make contact after reaching maturity.

Cell-mediated immunity protects against intracellular organisms

• Another class of lymphocyte, the T-cell, is concerned with control of intracellular infections. Like the B-cell, each T-cell has its individual antigen receptor (although it differs structurally from antibody) which recognizes antigen and the cell then undergoes clonal expansion to form effector and memory cells providing specific acquired immunity.

• The T-cell recognizes cell surface antigens in association with molecules of the MHC. Naive T-cells are only stimulated to undergo a primary response by specialized dendritic antigen-presenting cells.

• Primed T-helper cells, which see antigen with class II MHC on the surface of macrophages, release cytokines which in some cases can help B-cells to make antibody and in others activate macrophages and enable them to kill intracellular parasites.

• Cytotoxic T-cells have the ability to recognize specific antigen plus class I MHC on the surface of virally infected cells which are killed before the virus replicates. They also release γ-interferon which can make surrounding cells resistant to viral spread (figure 2.19).

• NK cells have lectin-like 'nonspecific' receptors for cells infected by viruses but do not have antigen-specific receptors; however, they can recognize antibody-coated virally infected cells through their Fcγ receptors and kill the target by antibody-dependent cellular cytotoxicity (ADCC).

• Although the innate mechanisms do not improve with repeated exposure to infection as do the acquired, they play a vital role since they are intimately linked to the acquired systems by **two different pathways** which all but **encapsulate the whole of immunology**. Antibody, complement and polymorphs give protection against most extracellular organisms, while T-cells, soluble cytokines, macrophages and NK cells deal with intracellular infections (figure 2.20).

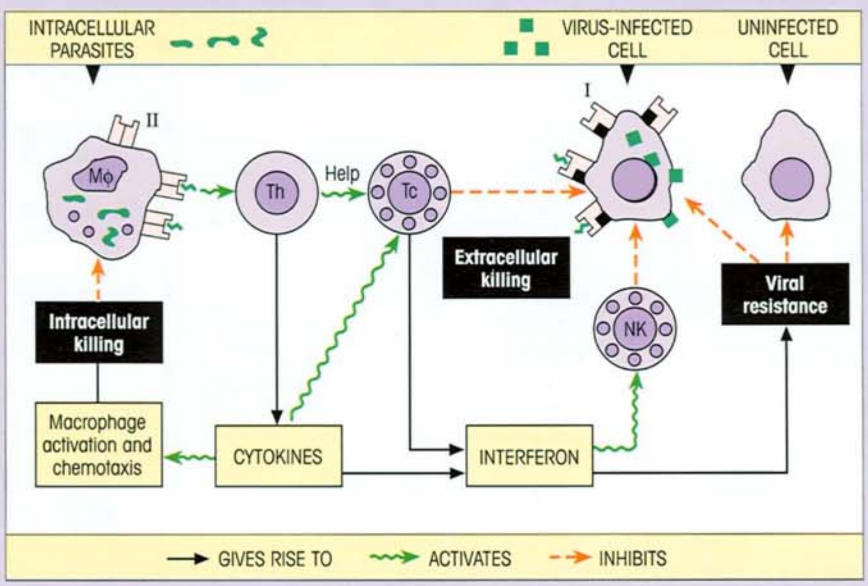


Figure 2.19. T-cells link with the innate immune system to resist intracellular infection. Class I ( ) and class II ( ) major histocompatibility molecules are important for T-cell recognition of surface antigen. The T-helper cells (Th) cooperate in the development of cytotoxic T-cells (Tc) from precursors. The macrophage (Mφ) microbicidal mechanisms are switched on by macrophage activating lymphokines. Interferon inhibits viral replication and stimulates NK cells which, together with Tc, kill virus-infected cells.

(continued)

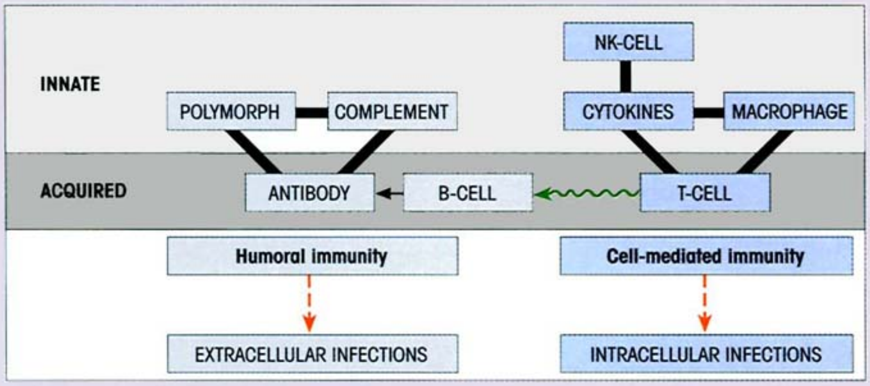


Figure 2.20. The two pathways linking innate and acquired immunity which provide the basis for humoral and cell-mediated immunity, respectively.

Immunopathology

- Immunopathologically mediated tissue damage to the host can occur as a result of:

inappropriate hypersensitivity reactions to exogenous antigens;

loss of tolerance to self giving rise to autoimmune disease;

reaction to foreign grafts.

- Immunodeficiency leaves the individual susceptible to infection.

See the accompanying website (**www.roitt.com**) for multiple choice questions.

FURTHER READING

General reading

Aderem A. & Underhill D.M. (1999) Mechanisms of phagocytosis in macrophages. *Annual Review of Immunology* **17**, 593.

Alt F. & Marrack P. (eds) (2001) *Current Opinion in Immunology* **13**. (Appears bimonthly; Issue No. 1 deals with 'Innate Immunity'.)

Gudmundsson G.H. & Agerberth B. (1999) Neutrophil antibacterial peptides, multifunctional effector molecules in the mammalian immune system. *Journal of Immunological Methods* **232**, 45.

Levy O. (1996) Antibiotic proteins of polymorphonuclear leukocytes. *European Journal of Haematology* **56**, 263.

Matzinger P. (1998) An innate sense of danger. *Seminars in Immunology* **10**, 399.

Morgan B.P. (1991) *Complement. Clinical Aspects and Relevance to Disease.* Academic Press, London.

Neth O. *et al.* (2000) Mannose-binding lectin binds to a range of clinically relevant micro-organisms and promotes complement deposition. *Infection and Immunity* **68**, 688.

Reid K.B.M. (1995) The complement system—a major effector mechanism in humoral immunity. *The Immunologist* **3**, 206.

Segal A.W. & Abo A. (1993) The biochemical basis of the NADPH oxidase of phagocytes. *Trends in Biochemical Sciences* **18**, 43.

Shi L., Krant R.P., Aebersold R. & Greenberg A.H. (1992) An NK cell granule protein that induces DNA fragmentation and apoptosis (in perforin permeabilized cells). *Journal of Experimental Medicine* **175**, 553.

Sitaram N. & Nagaraj R. (1999) Interaction of antimicrobial peptides with biological and model membranes: structural and charge requirements for activity. *Biochimica et Biophysica Acta—Biomembranes* **1462**, 29.

Reference work

Delves P.J. & Roitt I.M. (eds) (1998) *Encyclopedia of Immunology*, 2nd edn. Academic Press, London. (Covers virtually all aspects of the subject and describes immune responses to most infections.)

Historical

Clarke W.R. (1991) *The Experimental Foundations of Modern Immunology*, 4th edn. John Wiley & Sons, New York. (Important for those wishing to appreciate the experiments leading up to many of the major discoveries.)

Ehrlich P. (1890) On immunity with special reference to cell life. In Melchers F. *et al.* (eds) *Progress in Immunology* **VII**. Springer-Verlag, Berlin. (Translation of a lecture to the Royal Society (London) on the side-chain theory of antibody formation, showing this man's perceptive genius. A must!)

Landsteiner K. (1946) *The Specificity of Serological Reactions*. Harvard University Press (reprinted 1962 by Dover Publications, New York).

Mazumdar P.M.M. (ed.) (1989) *Immunology 1930–1980*. Wall & Thompson, Toronto.

Metchnikoff E. (1893) *Comparative Pathology of Inflammation*. Kegan Paul, Trench, Trubner, London (translated by F.A. & E.H. Starling).

Palmer R. (ed.) (1993) *Outstanding Papers in Biology*. Current Biology, London. (A delight to browse through some of the seminal papers which have shaped modern biology; wonderful material for teaching.)

Silverstein A.M. (1989) *A History of Immunology*. Academic Press, San Diego.

Tauber A.I. (1991) *Metchnikoff and the Origins of Immunology*. Oxford University Press, Oxford.

Table 2.1. The major immunologic journals and their impact factors.

GENERAL JOURNALS OF PARTICULAR INTEREST TO IMMUNOLOGISTS	*IMPACT FACTOR
Cell	36.2
EMBO J.	14.0
Lancet	10.2
Nature	29.5
Nature Medicine	26.6
New England Journal of Medicine	28.9
Proceedings of the National Academy of Sciences of the USA	10.3
Science	24.6
IMMUNOLOGICAL JOURNALS	***IMPACT FACTOR**
Autoimmunity	1.1
Cancer Immunology Immunotherapy	2.3
Cellular Immunology	2.3
Clinical and Experimental Allergy	2.7
Clinical and Experimental Immunology	2.8
European Journal of Immunology	5.6
Human Immunology	2.6
Immunity	20.6
Immunogenetics	2.9
Immunology	2.6
Immunopharmacology	1.4
Infection and Immunity	4.2
International Archives of Allergy and Applied Immunology	1.9
International Immunology	2.9
Journal of Allergy and Clinical Immunology	4.6
Journal of Autoimmunity	2.2
Journal of Clinical Immunology	2.4
Journal of Experimental Medicine	15.7
Journal of Immunology	7.1
Journal of Immunological Methods	1.9
Journal of Immunotherapy	3.1
Journal of Reproductive Immunology	1.5
Molecular Immunology	2.1
Nature Immunology	N/A
Parasite Immunology	2.0
Scandinavian Journal of Immunology	1.7
Tissue Antigens	3.0
Transplantaion	3.5
Vaccine	3.2

***IMPACT FACTOR** = relative frequency with which the journal's 'average article' has been cited in other publications

N/A, not available.

In-depth series for the advanced reader

Advances in Immunology (Annual). Academic Press, London.

Advances in Neuroimmunology (edited by G.B. Stefano & E.M. Smith). Pergamon, Oxford.

Annual Review of Immunology. Annual Reviews Inc., California.

Immunological Reviews (edited by P. Parham). Munksgaard, Copenhagen. (Specialized, authoritative and thoughtful.)

Progress in Allergy. Karger, Basle.

Seminars in Immunology. Academic Press, Cambridge. (In-depth treatment of single subjects.)

Current information

Current Biology. Current Biology, London. (What the complete biologist needs to know about significant current advances.)

Current Opinion in Immunology. Current Science, London. (Important personal opinions on focused highlights of the advances made in the previous year; most valuable for the serious immunologist.)

Trends in Molecular Medicine. Elsevier Science Publications, Amsterdam. (Frequent articles of interest to immunologists with very good perspective.)

The Immunologist. Hogrefe & Huber Publishers, Seattle. (Official organ of the International Union of Immunological Societies—IUIS. Excellent, didactic and compact articles on current trends in immunology.)

Trends in Immunology. Elsevier Science Publications, Amsterdam. (The immunologist's 'newspaper'. Excellent.)

Multiple choice questions

Roitt I.M. & Delves P.J. 400 MCQs each with five annotated learning responses. See **Website**.

Website (linked to Roitt's Essential Immunology*)*

http://www.roitt.com

The Website contains (i) 400 *Multiple Choice Revision Questions*, (ii) continual *update* providing abstracts and comments on important current immunologic publications, (iii) a series of *Interactive Core Tutorials in Immunology*. Blackwell Science Ltd, Oxford (CD ROM Mac/Windows: 110 fundamental immunologic principles using animations accompanied by spoken explanations to clarify and guide. Reference to standard and advanced text in *Essential Immunology* and a selection of multiple choice questions), (iv) *Further Reading* and reference archive, and (v) *Image archive* with over 400 illustrations from *Essential Immunology*.

Major journals

The major journals of interest and their impact factors are noted in table 2.1.

CHAPTER 3

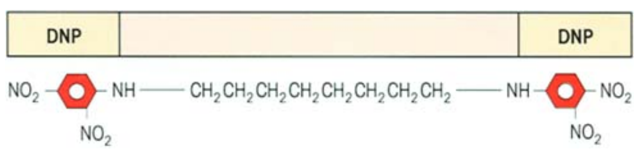
Antibodies

INTRODUCTION

The antibody molecule is made up of two identical heavy and two identical light chains held together by interchain disulfide bonds (figure M3.1.1). These chains can be separated by reduction of the S–S bonds and acidification. In the most abundant type of antibody in the circulation, **immunoglobulin G**, the exposed hinge region is extended in structure due to the high proline content and is therefore vulnerable to proteolytic attack; thus the molecule can be easily split in the laboratory using papain to yield two identical **Fab** fragments, each with a single combining site for antigen, and a third fragment, **Fc**, which lacks the ability to bind antigen. Pepsin strikes at a different point and cleaves the Fc from the remainder of the molecule to leave a large 5*S* fragment which is designated $F(ab')_2$ since it is still divalent with respect to antigen binding just like the parent antibody (Milestone 3.1).

ANTIBODIES HAVE DISCREET SITES FOR BINDING ANTIGEN

The location of the antigen combining sites was elegantly demonstrated by a study of purified antibodies to the dinitrophenyl (DNP) group mixed with the compound:



The two DNP groups are far enough apart not to interfere with each other's combination with antibody so that they can bring the antigen combining sites on two different antibodies together end to end. When viewed by negative staining in the electron microscope, a series of geometric forms is observed which represents the different structures to be expected if a flexible Y-shaped hinged molecule with a combining site at the end of each of the two arms of the Y were to complex with this divalent antigen. Triangular trimers, square tetramers and pentagonal pentamers may be readily discerned (figure 3.1). The way in which these polymeric forms arise is indicated in figure 3.2. The position of the Fc fragment and its lack of involvement in the combination with antigen are apparent from the shape of the polymers formed using the pepsin $F(ab')_2$ fragment (figure 3.1C).

AMINO ACID SEQUENCES REVEAL VARIATIONS IN IMMUNOGLOBULIN STRUCTURE

For good reasons, the antibody population in any

Milestone 3.1 —Four-polypeptide Structure of Immunoglobulin Monomers

Early studies showed the bulk of the antibody activity in serum to be in the slow electrophoretic fraction termed γ-globulin (subsequently immunoglobulin). The most abundant antibodies were divalent, i.e. had two combining sites for antigen and could thus form a precipitating complex (cf. figure 6.2).

To Rodney Porter and Gerald Edelman must go the credit for unlocking the secrets of the basic structure of the immunoglobulin molecule. If the internal disulfide bonds are reduced, the component polypeptide chains still hang together by strong noncovalent attractions. However, if the reduced molecule is held under acid conditions, these attractive forces are lost as the chains become positively charged and can now be separated by gel filtration into larger so-called heavy chains of approximately 55 000 Da (for IgG, IgA and IgD) or 70 000 Da (for IgM and IgE) and smaller light chains of about 24 000 Da.

The clues to how the chains are assembled to form the antibody molecule came from selective cleavage using proteolytic enzymes. Papain destroyed the precipitating power of the intact molecule but produced two univalent Fab fragments still capable of binding to antigen (Fab = *fragment antigen binding*); the remaining fragment had no affinity for antigen and was termed Fc by Porter (*fragment crystallizable*). After digestion with pepsin a molecule called $F(ab')_2$ was isolated; it still precipitated antigen and so retained both binding sites, but the Fc portion was further degraded. The structural basis for these observations is clearly evident from figure M3.1.1. In essence, with minor changes, all immunoglobulin molecules are constructed from one or more of the basic four-chain monomer units.

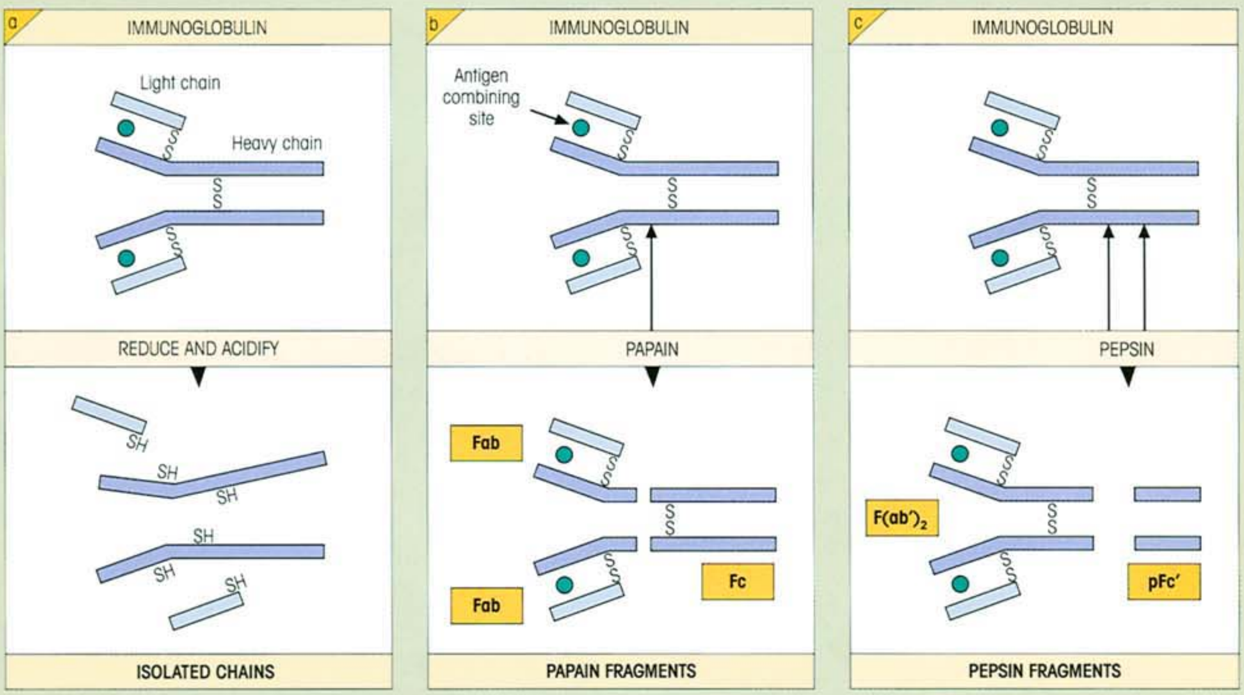


Figure M3.1.1. The antibody basic unit, consisting of two identical heavy and two identical light chains held together by interchain disulfide bonds, can be broken down into its constituent polypeptide chains and to proteolytic fragments, the pepsin $F(ab')_2$ retaining two binding sites for antigen and the papain Fab with one. After pepsin digestion the pFc′ fragment representing the C-terminal half of the Fc region is formed and is held together by noncovalent bonds. The portion of the heavy chain in the Fab fragment is given the symbol Fd. The N-terminal residue is on the left for each chain.

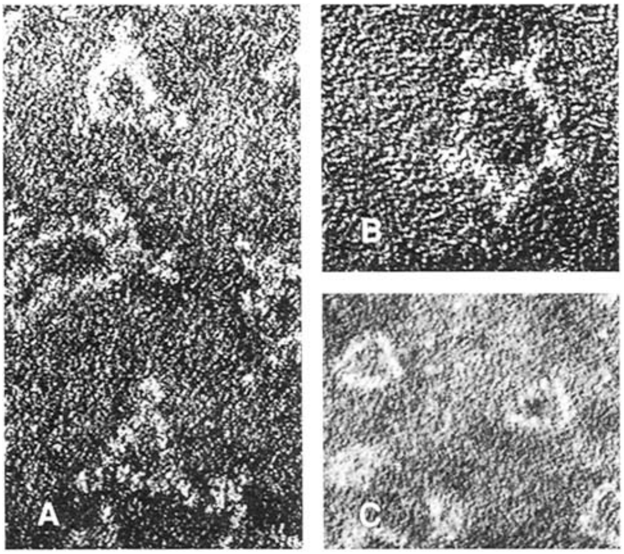


Figure 3.1. (A) and (B) Electron micrograph (×1 000 000) of complexes formed on mixing the divalent DNP hapten with rabbit anti-DNP antibodies. The 'negative stain' phosphotungstic acid is an electron-dense solution which penetrates into the spaces between the protein molecules. Thus the protein stands out as a 'light' structure in the electron beam. The hapten links together the Y-shaped antibody molecules to form trimers (A) and pentamers (B) (cf. figure 3.2). The flexibility of the molecule at the hinge region is evident from the variation in angle of the arms of the 'Y'. (C) As in (A); but the trimers were formed using the $F(ab')_2$ antibody fragment from which the Fc structures have been digested by pepsin (× 500 000). The trimers can be seen to lack the Fc projections at each corner evident in (A). (After Valentine R.C. & Green N.M. (1967) *Journal of Molecular Biology* **27**, 615; courtesy of Dr Green and with the permission of Academic Press, New York.)

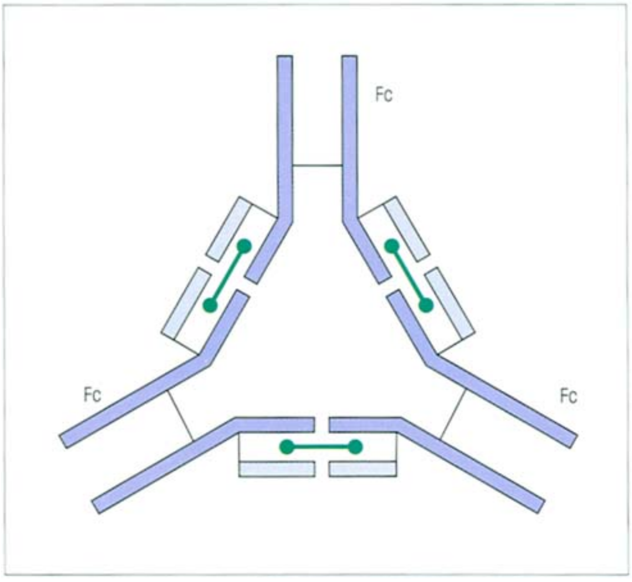


Figure 3.2. Three DNP antibody molecules held together as a trimer by the divalent antigen (●——●). Compare figure 3.1A. When the Fc fragments are first removed by pepsin, the corner pieces are no longer visible (figure 3.1C).

given individual is just incredibly heterogeneous, and this has meant that the determination of amino acid sequences was utterly useless until it proved possible to obtain the homogeneous product of a single clone. The opportunity to do this first came from the study of **myeloma proteins**.

In the human disease known as multiple myeloma, one cell making one particular individual immunoglobulin divides over and over again in the uncontrolled way a cancer cell does, without regard for the overall requirement of the host. The patient then possesses enormous numbers of identical cells derived as a clone from the original cell and they all synthesize the same immunoglobulin—the myeloma or M-protein—which appears in the serum, sometimes in very high concentrations. By purification of the myeloma protein we can obtain a preparation of an immunoglobulin having a unique structure. **Monoclonal antibodies** can also be obtained by fusing individual antibody-forming cells with a B-cell tumor to produce a constantly dividing clone of cells dedicated to making the one antibody (cf. figures 2.11 and 6.20).

The sequencing of a number of such proteins has revealed that the N-terminal portions of both heavy and light chains show considerable variability, whereas the remaining parts of the chains are relatively constant, being grouped into a restricted number of structures. It is conventional to speak of variable and constant regions of both heavy and light chains (figure 3.3).

Certain sequences in the variable regions show quite remarkable diversity and systematic analysis localizes these hypervariable sequences to three segments on the light chain (figure 3.4) and three on the heavy chain.

IMMUNOGLOBULIN GENES

Immunoglobulins are encoded by multiple gene segments

Clusters of genes on three different chromosomes code for κ light chains, λ light chains and for heavy chains, respectively. Since a wide range of antibodies with differing amino acid sequences can be produced, there must be corresponding nucleotide sequences to encode them. However, the complete gene encoding each heavy and light chain is not present as such in the inherited (germ-line) DNA, but is created during early development of the B-cell by the joining together of minisegments of the gene. Take the human κ light chain for example; the variable region is made up of two gene

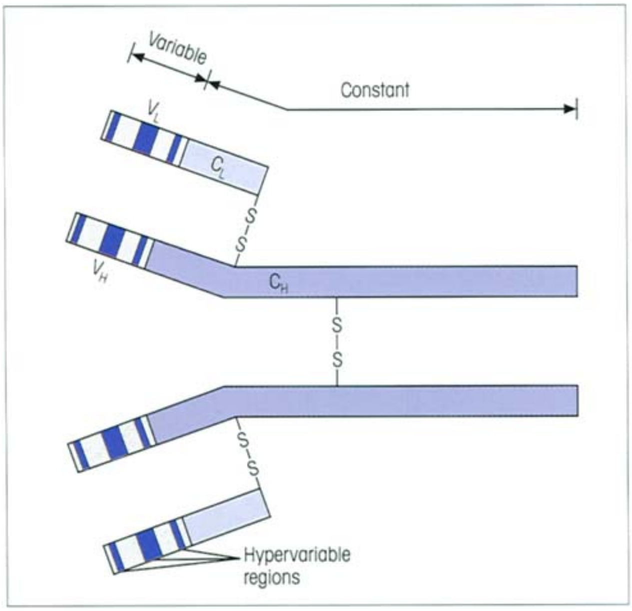


Figure 3.3. Amino acid sequence variability in the antibody molecule. The terms 'V region' and 'C region' are used to designate the variable and constant regions, respectively; 'V_L' and 'C_L' are generic terms for these regions on the light chain and 'V_H' and 'C_H' specify variable and constant regions on the heavy chain. Certain segments of the variable region are hypervariable but adjacent framework regions are more conserved. As stressed previously, each pair of heavy chains is identical, as is each pair of light chains.

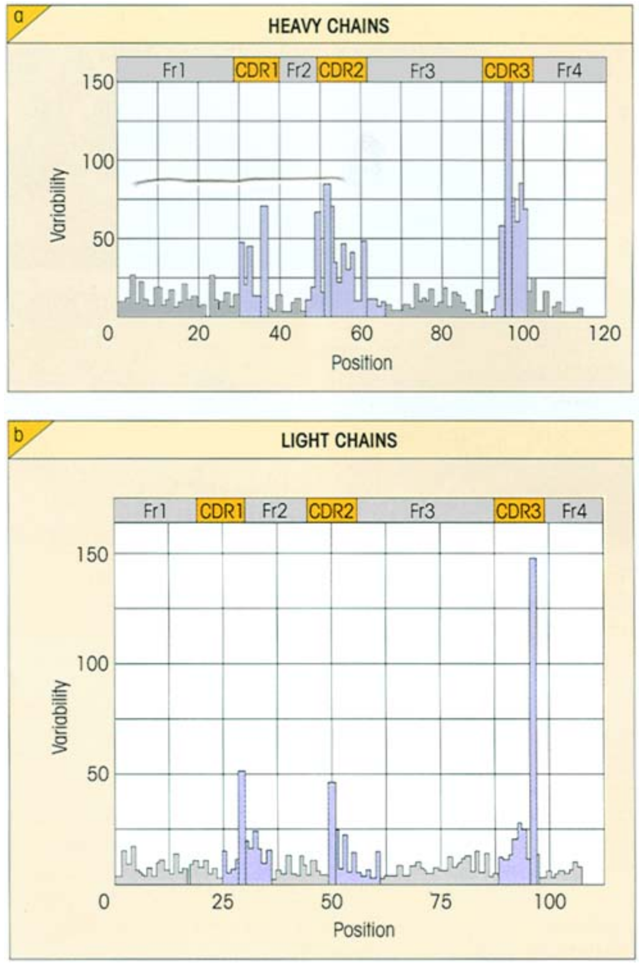


Figure 3.4. Wu and Kabat plot of amino acid variability in the variable region of immunoglobulin heavy and light chains. The sequences of chains from a large number of myeloma monoclonal proteins are compared and the variability at each position is computed as the number of different amino acids found divided by the frequency of the most common amino acid. Obviously, the higher the number the greater the variability; for a residue at which all 20 amino acids occur randomly, the number will be 400 ($20 \div 0.05$), and at a completely invariant residue, the figure will be 1 ($1 \div 1$). The three hypervariable regions (blue) in the (a) heavy and (b) light chains, usually referred to as **complementarity determining regions (CDRs)**, are clearly defined. The intervening peptide sequences (gray) are termed framework regions (Fr1–4). (Courtesy of Professor E.A. Kabat.)

segments, a large V_κ and a small J_κ, while a single gene encodes the constant region (figure 3.5). There is a cluster of ~40 V_κ genes and just five functional J_κ genes. In the immature B-cell, a translocation event leads to the joining of one of the V_κ genes to one of the J_κ segments. Each V segment has its own leader sequence and a number of upstream promoter sites, including a characteristic octamer sequence, to which transcription factors bind (figure 3.6). When the Ig gene is transcribed, splicing of the nuclear RNA brings the $V_\kappa J_\kappa$ sequence into contiguity with the constant region C_κ sequence, the whole being read off as a continuous κ chain polypeptide within the endoplasmic reticulum.

The same general principles apply to the arrangement of λ light chain and the heavy chain genes, although the latter constellation shows additional features: the different constant region genes form a single cluster and there is a group of around 25 highly variable D segments located between the V and J regions (figure 3.7). The D segment together with its junctions to the V and J segments encodes almost the entire third hypervariable region, the first two hypervariable regions being encoded entirely within the V sequence.

A special mechanism effects *VDJ* recombination

In essence, the translocation involves the mutual recognition of conserved heptamer–spacer–nonamer recombination signal sequences which flank each germ-line V, D and J segment (figure 3.8). The products of the recombination activating genes *RAG-1* and *RAG-2* catalyse the introduction of double-strand breaks between the elements to be joined and their respective flanking sequences. At this stage, nucleotides may either be deleted or inserted between the *VD*, *DJ* or *VJ* joining elements before they are ultimately ligated.

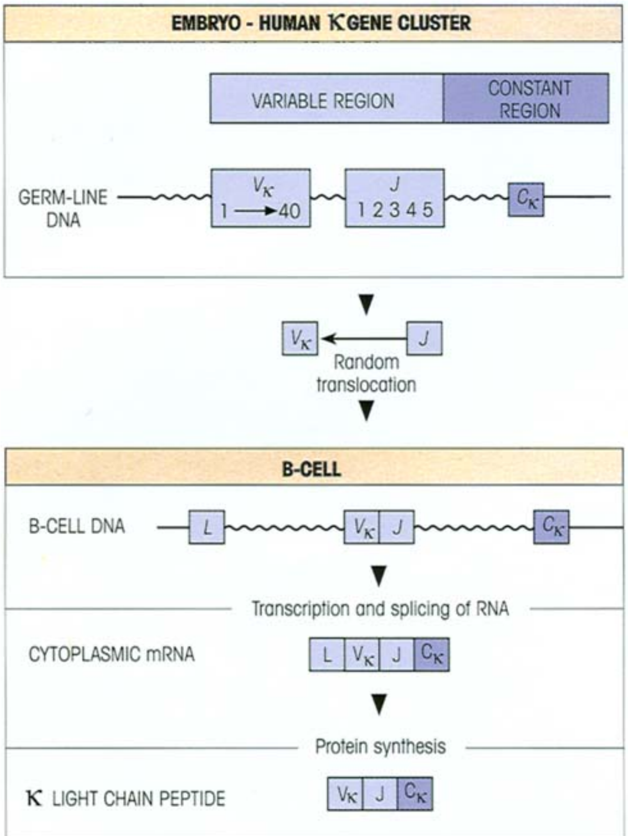


Figure 3.5. Genetic basis for synthesis of human κ chains. The V_κ genes are arranged in a series of families or sets of a closely related sequence. Each V_κ gene has its own leader sequence (*L*). As the cell becomes immunocompetent, the variable region is formed by the random combination of one out of the 40 V_κ genes with one of the five joining segments *J*, a gene rearrangement process facilitated by base sequences in the intron following the 3′ end of the V_κ segment pairing up with sequences in the intron 5′ to *J*. The final joining occurs when the intervening intron sequence between *J* and C_κ is spliced out of the RNA transcript. By convention, the genes are represented in italics and the antigens they encode in normal type.

STRUCTURAL VARIANTS OF THE BASIC IMMUNOGLOBULIN MOLECULE

Isotypes

Based upon the structure of their heavy chain constant regions, immunoglobulins are classed into major groups termed **classes** which may be further subdivided into **subclasses**. In the human, for example, there are five classes: immunoglobulin G (IgG), IgA, IgM, IgD and IgE. They may be differentiated not only by their sequences but also by the antigenic structures to which these sequences give rise. Thus, by injecting a human IgG myeloma protein into a rabbit, it is possible to raise an antiserum which can be absorbed by mixtures of myelomas of other classes to remove cross-reacting antibodies and which will then be capable of reacting with IgG, but not IgA, IgM, IgD or IgE (figure 3.9).

Since all the heavy chain constant region (C_H) structures which give rise to classes and subclasses are expressed together in the serum of a normal subject, they are termed **isotypic variants** (table 3.1). Likewise, the light chain constant regions (C_L) exist in isotypic forms known as κ and λ which are associated with all heavy chain isotypes. Because the light chains in a given antibody are identical, immunoglobulins are either κ or λ but never mixed. Thus IgG exists as IgGκ or IgGλ, IgM as IgMκ or IgMλ, and so on.

Allotypes

This type of variation depends upon the existence of allelic forms (encoded by alleles or alternative genes at a single locus) which therefore provide genetic markers (table 3.1). In somewhat the same way as the red cells in genetically different individuals can differ in terms of the blood group antigen system A, B, O, so the Ig heavy chains differ in the expression of their allotypic groups. Typical allotypes are the **Gm specificities** on IgG

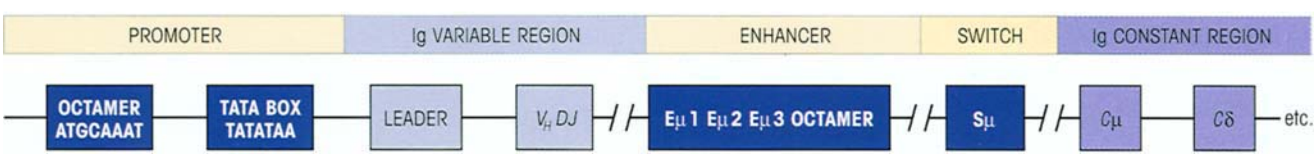


Figure 3.6. Genes controlling immunoglobulin heavy chain transcription. Each *VDJ* segment encoding the variable region is associated with a leader sequence. Closely upstream is the TATA box of the promoter which binds RNA polymerase II and the octamer motif which is one of a number of short sequences which bind transacting regulatory transcription factors. The *V*-region promoters are relatively inactive and only association with enhancers, which are also composites of short sequence motifs capable of binding nuclear proteins, will increase the transcription rate to levels typical of actively secreting B-cells. The enhancers are near to the regions which control switching from one Ig class constant region to another, e.g. IgM to IgG. Primary transcripts are initiated 20 nucleotides downstream of the TATA box and extend beyond the end of the constant region. These are spliced, cleaved at the 3′ end and polyadenylated to generate the translatable mRNA.

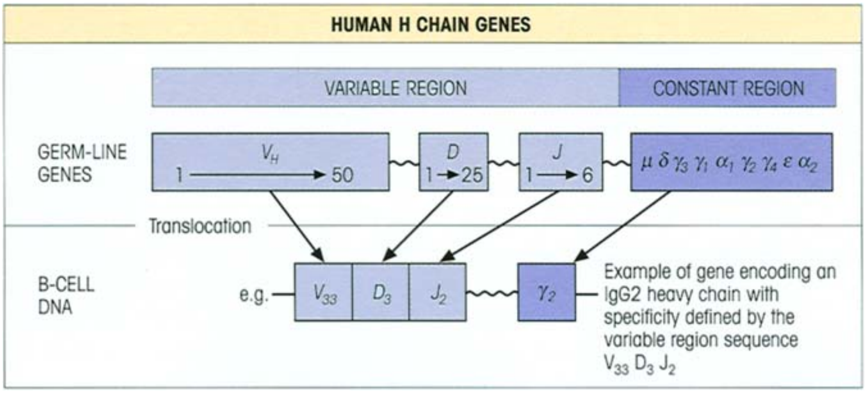
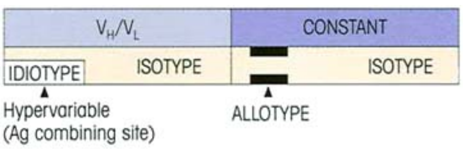


Figure 3.7. Human *V*-region genes shuffled by gene rearrangement to generate the single heavy chain specificity characteristic of each B-cell. The V_H genes can be grouped into seven different families (V_H1–V_H7) based upon sequence homology of > 80%, with V_H3 being by far the largest family. Individual members of a family are interspersed throughout the locus, i.e. there is no significant grouping together of V_H family genes. Note the additional *D* segment minigenes that are present in the heavy chain locus.

Table 3.1. Summary of immunoglobulin variants.



TYPE OF VARIATION	DISTRIBUTION	VARIANT	LOCATION	EXAMPLES
ISOTYPIC	All variants present in serum of a normal individual	Classes Subclasses Types Subgroups Families	C_H C_H C_L C_L V_H/V_L	IgM, IgE IgA1, IgA2 κ,λ λOz^+, λOz^- $V_{\kappa I}$ $V_{\kappa II}$ $V_{\kappa III}$ V_{HI} V_{HII} V_{HIII}
ALLOTYPIC	Alternative forms: genetically controlled so not present in all individuals	Allotypes	Mainly C_H/C_L sometimes V_H/V_L	Gm groups (human) b4, b5, b6, b9 (rabbit light chains) Igh-1^a, Igh-1^b (mouse γ_{2a} heavy chains)
IDIOTYPIC	Individually specific to each immunoglobulin molecule	Idiotypes	Variable regions	One or more hypervariable regions forming the antigen-combining site

(Gm = *marker* on IgG). Allotypic differences at a given Gm locus usually involve one or two amino acids in the peptide chain. Take, for example, the G1m(a) locus on IgG1 (table 3.1). An individual with this allotype would have the peptide sequence Asp.Glu.Leu.Thr.Lys on each of his or her IgG1 molecules. An individual possessing the a-negative allotype would have the sequence Glu.Glu.Met.Thr.Lys, i.e. two amino acids different. To date, 20 Gm groups have been found on the γ heavy chains and a further three (the Km—previously Inv groups) on the κ constant region.

As in other allelic systems, individuals may be homozygous or heterozygous for the genes encoding the markers; these are expressed codominantly and are inherited in simple Mendelian fashion. Take, for example, the b4, b5 allotypes on rabbit light chains: an animal of b^4b^4 genotype would express the b4 allotype, whereas a rabbit of b^4b^5 genotype derived from b^4b^4 and b^5b^5 parents would express the b4 marker on one fraction and b5 on another fraction of its immunoglobulin molecules.

Idiotypes

We have seen that it is possible to obtain antibodies that recognize isotypic and allotypic variants; one can also raise antiserums which are specific for individual antibody molecules and discriminate between one monoclonal antibody and another independently of isotypic or allotypic structures (figure 3.9). Such antiserums define the individual determinants character-

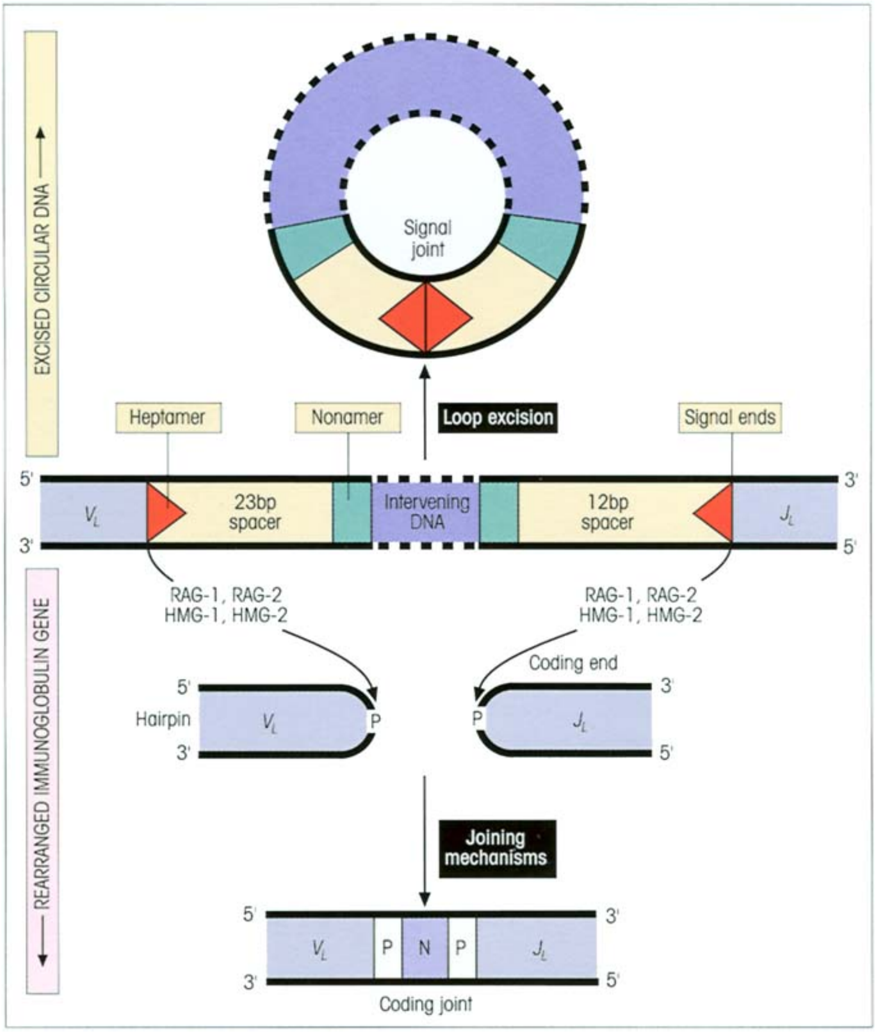


Figure 3.8. The joining of *V*, *D* and *J* segments. Joining is masterminded by the recombination activating genes *RAG-1* and *RAG-2*, the products of which cleave the DNA at the signal ends. RAG-1 and RAG-2 together produce several thousand times more efficient *VDJ* recombination than either alone. The introns adjoining the *V*, *D* and *J* gene segments contain specialized **recombination signal sequences (RSSs)** which include conserved heptamers and nonamers separated by spacers of either 12 or 23 base pairs. The two joining segments, in this example V_L and J_L, are brought into proximity by interaction between their respective RSSs mediated by the DNA bending and looping high mobility group-1 and -2 proteins (HMG-1 and HMG-2). RAG-1 and RAG-2 cleave the DNA to produce double-strand breaks at the border of the RSS. The excised signal sequences are ligated to form the signal joint resulting in a piece of circular DNA containing the excised sequences. This is probably maintained in the cell for a period of time before eventually being lost from the cell. The double strands of each coding segment form 'hairpin' ends. The enzyme Ku (a dimer of Ku70 and Ku86) binds to the DNA ends and stimulates DNA-dependent protein kinase (DNA-PK, mutation of which gives rise to mice with severe combined immunodeficiency (SCID)) which facilitates the opening of the hairpin. Terminal deoxynucleotidyl transferase (TdT) adds nucleotides to the ends of the DNA strands in order to generate N-region diversity. Unlike the precision of the signal joint, the coding joint is variable because it can involve the addition of base pairs resulting from both the resolution of the hairpin loop (P-elements) and the TdT-mediated N-region diversity. Nucleases remove any excess nucleotides and polymerases fill in any gaps before the DNA ligase IV and XRCC4 enzymes carry out ligation of the two sequences. Since the coding elements are joined at random with respect to the reading frames, two out of three events have two coding elements out of frame. Although apparently wasteful, this is evolutionarily tolerated because it confers so much benefit in the form of antigen receptor diversity. *VDJ* recombination products define the major antigen-binding domains.

istic of each antibody, collectively termed the **idiotype** (Kunkel & Oudin). Not surprisingly, it turns out that the idiotypic determinants are located in the variable part of the antibody associated with the hypervariable regions (figure 3.10).

Anti-idiotypes which react with one antibody and no other are said to recognize **private** idiotypes and provide further support for the idea that each antibody has a unique structure. Frequently, antibody molecules of closely similar amino acid structure may, in ad-

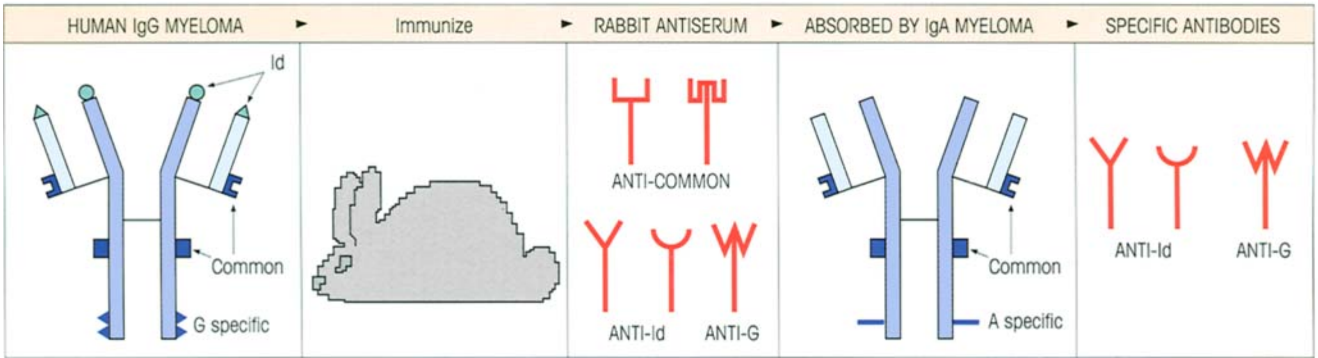


Figure 3.9. How to use monoclonal myeloma proteins to produce antibodies specific for different Ig structures. The rabbit makes antibodies directed to different parts of the human IgG myeloma. Antibodies to those parts which are common to other Ig classes can be absorbed out with myelomas of those classes leaving other antibodies reacting with class-specific G and variable region-specific (idiotype=Id) structures on the original molecule. By the same token, further absorption with other IgG myelomas will remove the common IgG-specific antibodies leaving an antiserum directed to the idiotypic determinants alone. (In an attempt to simplify, we have ignored subclasses and allotypes, but the same principles can be extended to the generation of antisera specific for these variants.) The rabbit produces a mixture of polyclonal antibodies directed against each structural site on the antigen, i.e. they are produced by clones derived from a variety of antigen-specific parent cells which each react stereochemically in a slightly different way with the same structure (cf. p. 62).

dition, share idiotypes (e.g. M104 and HDEX 2 in figure 3.10) and we then speak of **public** or **cross-reacting** idiotypes.

Anti-idiotypic serums provide useful reagents for demonstrating the same V region on different heavy chains and on different cells, for identification of specific immune complexes in patients' serums, for recognition of V_L type amyloid in subjects excreting

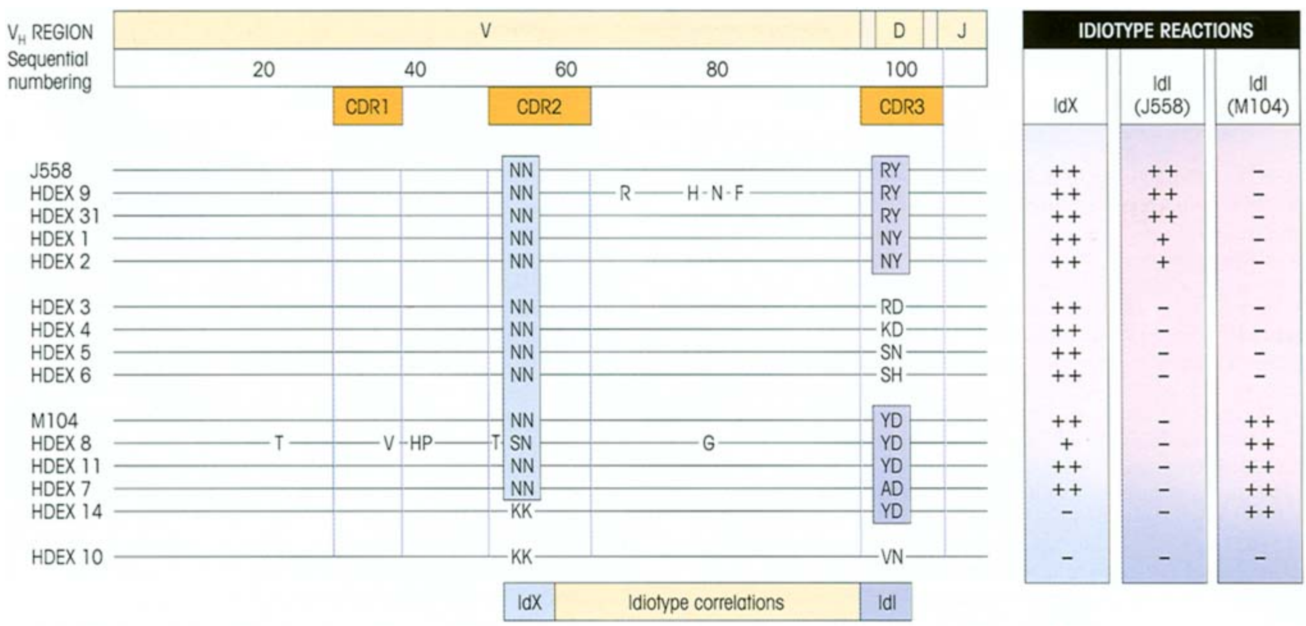


Figure 3.10. Structural correlates of idiotopes (individual determinants on an idiotype) on antidextran antibodies. Amino acid sequences of variable heavy chain regions of mouse monoclonal antidextran antibodies are shown. All antibodies have λ_1 L chains. Horizontal lines indicate identity to the sequence of the first protein, J558; letters (Dayhoff code) show differences or regions correlated with idiotopes (central boxed areas). The cross-reacting idiotope (IdX) is associated with the second complementarity determining region (CDR2) structures, while the private idiotopes (IdI) are features of the CDR3 region in these antibodies. The presence of the idiotopes on each antibody molecule is assessed by reaction with antisera specific for IdX, J558 IdI and M104 IdI (on the right). Cross-reacting idiotopes may also be associated with the CDR3 region in other systems. The hatched boxes either side of the D segment represent N-region diversity and P-elements. (Reproduced from Davie J.M. *et al.* (1986) *Annual Review of Immunology* **4**, 147 with permission. ©by Annual Reviews Inc.)

Bence-Jones proteins, for detection of residual monoclonal protein after therapy and perhaps for selecting lymphocytes with certain surface receptors. The reader will (or should) be startled to learn that it is possible to raise autoanti-idiotypic sera since this means that individuals can make antibodies to their own idiotypes. This has quite momentous consequences, as will become apparent when we discuss the Jerne network theory in Chapter 11.

IMMUNOGLOBULINS ARE FOLDED INTO GLOBULAR DOMAINS WHICH SUBSERVE DIFFERENT FUNCTIONS

Immunoglobulin domains have a characteristic structure

In addition to the *interchain* disulfide bonds shown by Edelman to link the heavy and light chains, there are internal *intrachain* disulfide bonds which stabilize each of the globular **immunoglobulin domains** (figure 3.11). Each domain has a characteristic β-pleated sheet structure and comprises approximately 110 amino acids, of which 65–70 are between the two cysteines which form the intrachain disulfide bond of each domain (see figure 3.13a). Significantly, the hypervariable sequences appear at one end of the variable domain where they form parts of the β-turn loops and are clustered close to each other in space.

The variable domain binds antigen

The clustering of the hypervariable loops at the tips of the variable regions where the antigen-binding site is localized makes them the obvious candidates to subserve the function of antigen recognition (figures 3.11 and 3.12). That this is indeed the case was confirmed when X-ray crystallographic analysis of complexes formed between the Fab fragments of monoclonal antibodies and their respective antigens was carried out. Indeed, the antigen specificity of a mouse monoclonal antibody can be conferred on a human immunoglobulin molecule by replacing the human hypervariable sequences with those of the mouse (cf. figure 6.21). The sequence heterogeneity of the three heavy and three light chain hypervariable loops ensures tremendous diversity in combining specificity for antigen through variation in the shape and nature of the surface they create. Thus each hypervariable region may be looked upon as an independent structure contributing to the complementarity of the binding site for antigen and one speaks of **complementarity determining regions (CDRs)**.

That these variable regions of heavy and light chains both contribute to antibody specificity is suggested by experiments in which isolated chains were examined for their antigen combining power. In general, varying degrees of residual activity were associated with the heavy chains but relatively little with the light chains; on recombination, however, there was always a significant increase in antigen-binding capacity. Interestingly though, the majority of immunoglobulins in camels and llamas lack light chains, although this does not seem to leave them unduly inconvenienced.

Constant region domains determine secondary biological function

The classes of antibody differ from each other in many respects: in their half-life, their distribution throughout the body, their ability to fix complement and their binding to cell surface Fc receptors. Since the classes all have the same κ and λ light chains, and heavy and light variable region domains, these differences must lie in the heavy chain constant regions.

These biological activities (effector functions) were originally localized to the various heavy chain domains by using myeloma proteins which have spontaneous domain deletions, or enzymic fragments produced by papain (which generates Fab and Fc fragments) or pepsin (which gives $F(ab')_2$ and pFc′, the C-terminal portion of Fc). More recently, the critical amino acids involved in individual functions have in many cases been identified using site-specific mutagenesis.

A model of the IgG molecule is presented in figure 3.13b which indicates the spatial disposition and interaction of the domains in IgG, while figure 3.13c ascribes the various biological functions to the relevant structures. In principle, the V-region domains form the recognition unit (cf. figure 2.1) and constant region domains mediate the secondary biological functions.

To enable the Fab arms to have the freedom to move and twist so that they can align their hypervariable regions with the antigenic sites on large immobile carriers, and to permit the Fc structures to adjust spatially in order to trigger their effector functions, it is desirable for IgG to have a high degree of flexibility. And it has just that (figure 3.13d). Structural analysis shows that the Fab can ‘elbow-bend’ at its V–C junction and twist about the hinge, which itself can more properly be described as a *loose tether*, allowing the Fab and the Fc to drift relative to each other with remarkable suppleness (cf. figure 3.1). It could be said that movements like that make it a very sexy molecule!

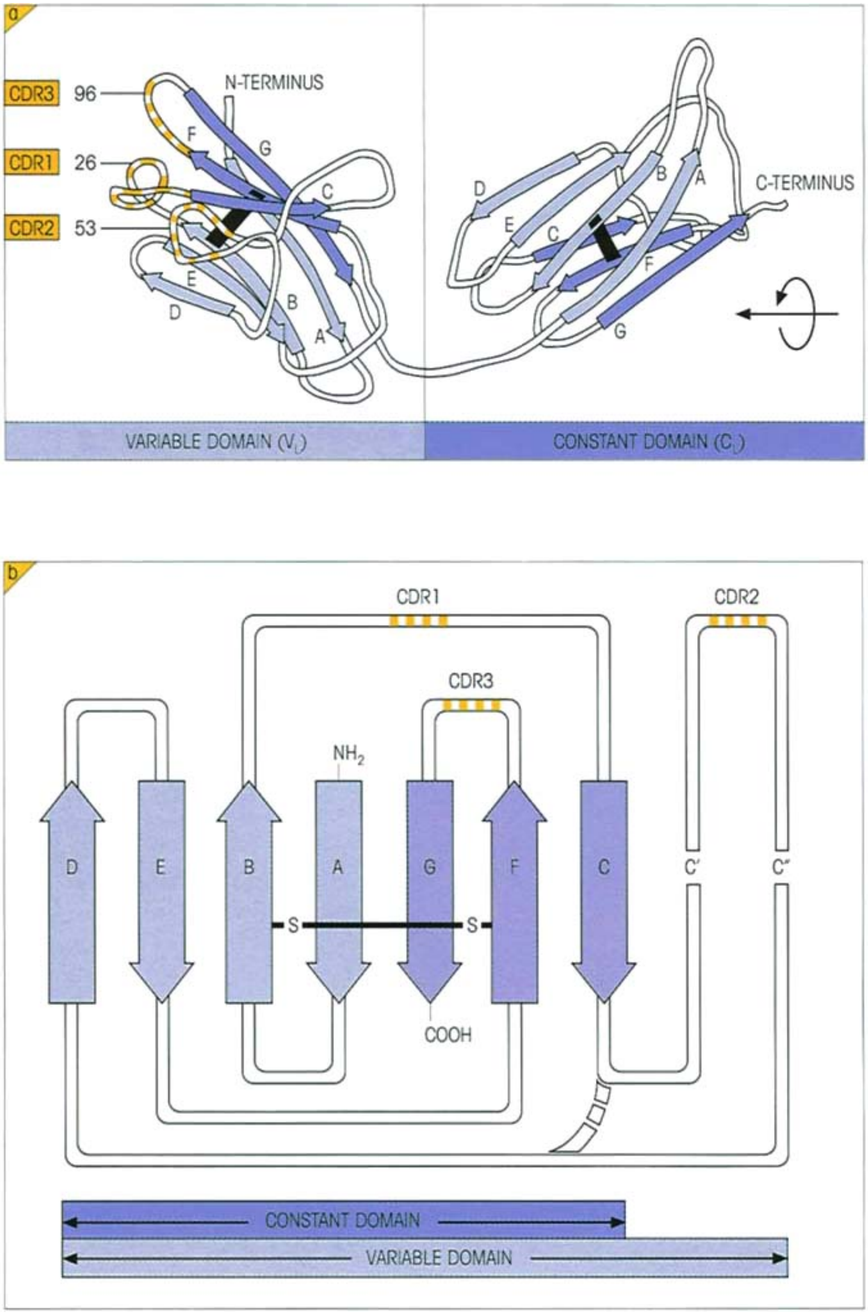


Figure 3.11. Ig domain structure. (a) Structure of the globular domains of a light chain (from X-ray crystallographic studies of a Bence-Jones protein by Schiffler *et al.* (1973) *Biochemistry* **12**, 4620). One surface of each domain is composed essentially of four strands (light blue arrows) arranged in an antiparallel β-pleated sheet structure stabilized by interstrand H bonds between the amide CO· and NH· groups running along the peptide backbone, and the other surface of a second β-pleated sheet of three strands (darker blue arrows); the black bar represents the intrachain disulfide bond. This structure is characteristic of all immunoglobulin domains. Of particular interest is the location of the hypervariable regions (▪ ▪ ▪ ▪) in three separate loops which are closely disposed relative to each other and form the light chain contribution to the antigen-binding site (cf. figure 3.12). One numbered residue from each complementarity determining region is identified. To generate a Fab fragment (cf. left side of figure 3.13d), imagine a V_H–C_H1 segment just like the V_L–C_L in the diagram, rotate it 180° around the axis of the arrow on the right of the figure and lay it on top of the V_L–C_L segment (Dr A. Feinstein). (b) Schematic view of folding pattern of constant and variable light chain domains showing the β-strands (A–G). The extra sequence C′C″ in the variable structure includes the CDR2 region and, in some variable region sequences, H bonding patterns occur which produce two extra β-strands, giving a five-stranded β-pleated sheet instead of a three-stranded one. Lettering and colors as in (a).

IMMUNOGLOBULIN CLASSES AND SUBCLASSES

The physical and biological characteristics of the five major immunoglobulin classes in the human are summarized in tables 3.2 and 3.3. The following comments are intended to supplement this information.

Immunoglobulin G has major but varied roles in extracellular defenses

Its relative abundance in the circulation and in most tissues, its ability to develop high affinity binding for antigen and its wide spectrum of secondary biological properties make IgG the prime workhorse of the Ig stable. IgG, together with IgA, is the major immunoglobulin to be synthesized during the secondary response. Because IgG diffuses

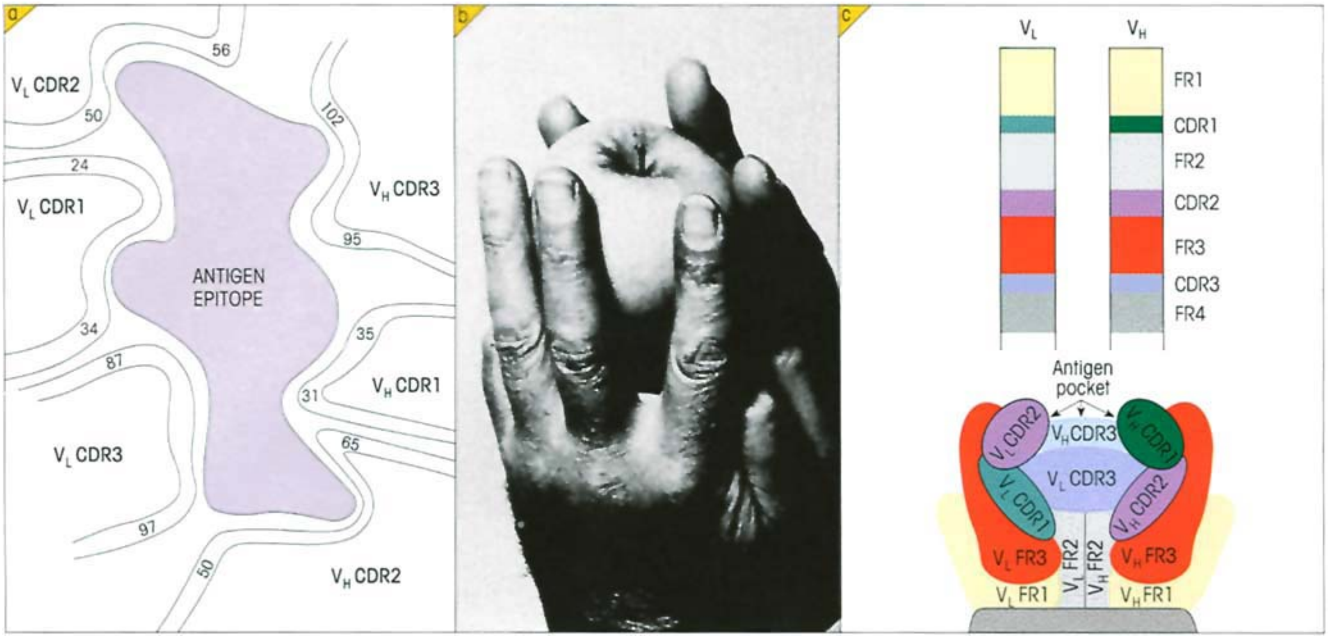


Figure 3.12. The binding site. (a) Idealized two-dimensional representation of an antigen-binding site formed by spatial apposition of polypeptide loops containing the hypervariable regions on light and heavy chains. Numbers refer to amino acid residues. In the heavy chain CDR1 is composed of amino acids 31–35, CDR2 50–65 and CDR3 95–102. For the light chain CDR1 is found at position 24–34, CDR2 at 50–56 and CDR3 at 87–97. Glycine residues are nearly always present at certain positions in or around the CDRs, allowing peptide chains to fold back and form β-pleated sheet structures which enable the hypervariable regions to lie close to each other (figure 3.11a ■). Wu and Kabat have suggested that the flexibility of the bond angle in this amino acid contributes to the effective formation of a binding site. By using different combinations of hypervariable regions and different residues within each of these regions, each antibody molecule may be able to form a complex with a variety of antigenic determinants (with a comparable variety of affinities). (b) A simulated combining site for a hapten formed by apposing the three middle fingers of each hand, each finger representing a hypervariable loop. With protein epitopes the area of contact is usually greater and tends to involve more superficial residues (cf. figure 5.5). There appears to be a small repertoire of main chain conformations for at least five of the six CDRs, the particular configuration adopted being determined by a few key conserved residues (Chothia C. *et al.* (1989) *Nature* **342**, 877). (Photograph by B.N.A. Rice; inspired by A. Munro!) (c) An idealized space-filling model indicating the relative locations of the six CDR loops (taken from Silverman G.J. (1994) *The Immunologist* **2**, 52, with permission.)

more readily than the other immunoglobulins into the extravascular body spaces, it is the predominant species in nonmucosal tissues, where it carries the major burden of neutralizing bacterial toxins and of binding to microorganisms to enhance their phagocytosis.

Activation of the classical complement pathway

Complexes of bacteria with IgG antibody trigger the C1 complex when a minimum of two Fcγ regions in the complex bind C1q (see figure 2.2). As shown by site-specific mutagenesis of the Cγ2 domain, the common core motif for binding to C1q is Glu.318-X-Lys.320-X-Lys.322 where X, the residue separating these charged amino acids, can vary. In keeping with this analysis, it is comforting to note that synthetic peptides with this structure can block C1q binding to IgG. Activation of the next component, C4, tends to produce attachment of C4b to the Cγ1 domain. Thereafter, one observes C3 convertase formation, covalent coupling of C3b to the bacteria and release of C3a and C5a leading to the chemotactic attraction of our friendly polymorphonuclear phagocytic cells (cf. p. 2). These adhere to the bacteria through surface receptors for complement and the Fc portion of IgG (Fcγ) and then ingest the microorganisms through phagocytosis. In a similar way, the extracellular killing of target cells coated with IgG antibody is mediated largely through recognition of the surface Fcγ by NK (natural killer) cells bearing the appropriate receptors (cf. p. 18). The thesis that the **biological individuality of different immunoglobulin classes is dependent on the heavy chain constant regions, particularly the Fc**, is amply borne out in relationship to activities such as transplacental passage, complement fixation and binding to various cell types, where function has been shown to be mediated by the Fc part of the molecule.

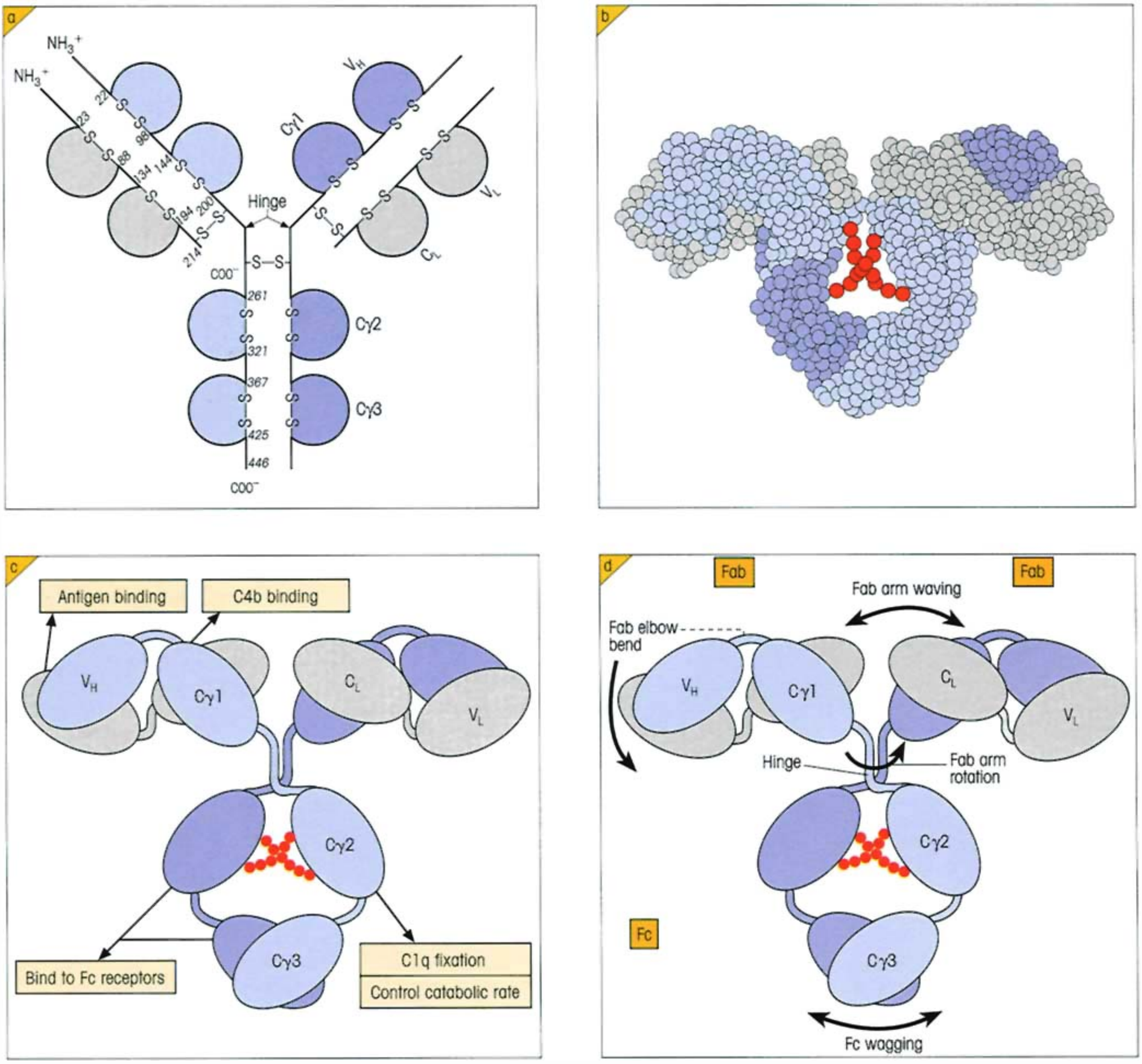


Figure 3.13. The disposition, interaction and biological properties of the Ig domains in IgG. (a) Stylized representation of IgG indicating typical positions of the intrachain and interchain disulfide bonds and the location of the hinge region. The number of inter-heavy chain disulfide bonds varies with the subclass of IgG; although for simplicity only one is shown, there are actually two in IgG1 and IgG4, four in IgG2 and a massive 11 in IgG3. (b) Computer-generated model of IgG. One heavy chain is depicted in light blue, the other in darker blue and the light chains in gray. Carbohydrate bound to and separating the Cγ2 domains is in red. The molecule analysed was a mutant with a truncated hinge region since the flexibility associated with the hinge (see (d) below) makes the X-ray pictures fuzzy and difficult to interpret. (The structure was determined by Silverton E.W., Navia M.A. & Davies J.R. (1977) *Proceedings of the National Academy of Sciences of the USA* **74**, 5140, and the figure generated by computer graphics using R.J. Feldmann's system (National Institute of Health).) (c) Diagram based on the model indicating domain location of biological function. The combined Cγ2 and Cγ3 domains bind to Fc receptors on phagocytic cells, NK cells and placental syncytiotrophoblast; also to staphylococcal protein A. (Note the IgG heavy chain is designated γ and the constant region domains Cγ1, Cγ2 and Cγ3.) (d) Diagram indicating IgG flexibility (cf. Brekke O.H. *et al.* (1995) *Immunology Today* **16**, 85) and showing apposing domains making contact through hydrophobic regions (after Dr A. Feinstein). The structures of these contact frameworks are highly conserved, an essential feature if different V_H and V_L domains are to associate in order to generate a wide variety of antibody specificities. These hydrophobic regions on the two complement-fixing C_H2 (Cγ2) domains are partly masked by carbohydrate and remain independent, so allowing the formation of a hinge region which is extremely flexible both with respect to variation in the angle of the Fab fragments (waving) and their rotation about the hinge peptide chain. This flexibility enables combining sites in IgG to be readily adapted to spatial variations in the presentation of the antigenic epitopes; Fc 'wagging' permits optimal orientation to receptors on effector cells.

Table 3.2. Physical properties of major human immunoglobulin classes.

DESIGNATION	IgG	IgA	IgM	IgD	IgE
Sedimentation coefficient	7S	7S,9S,11S*	19S	7S	8S
Molecular weight	150 000	160 000 and dimer	970 000	175 000	190 000
Number of basic four-peptide units	1	1,2*	5	1	1
Heavy chains	γ	α	μ	δ	ε
Light chains	κ or λ	κ or λ	κ or λ	κ or λ	κ or λ
Molecular formula†	$\gamma_2\kappa_2, \gamma_2\lambda_2$	$(\alpha_2\kappa_2)_{1-2}$ $(\alpha_2\lambda_2)_{1-2}$ $(\alpha_2\kappa_2)_2S^*$ $(\alpha_2\lambda_2)_2S^*$	$(\mu_2\kappa_2)_5$ $(\mu_2\lambda_2)_5$	$\delta_2\kappa_2$ $\delta_2\lambda_2$?	$\varepsilon_2\kappa_2, \varepsilon_2\lambda_2$
Valency for antigen binding	2	2,4	5(10)	2	2
Concentration range in normal serum	8-16 mg/ml	1.4-4 mg/ml	0.5-2 mg/ml	0.003-0.04 mg/ml	17-450 ng/ml‡
% Total immunoglobin	75	15	5-10	0-1	0.002
% Carbohydrate content	3	8	12	9	12

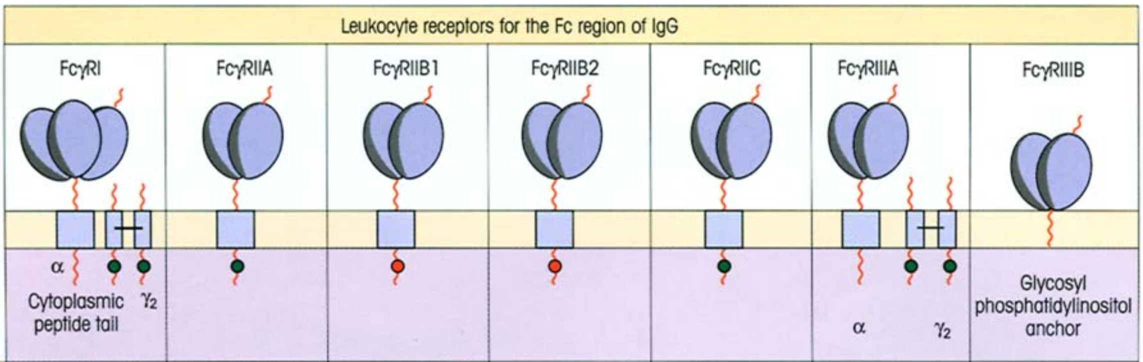
* 7S monomer, 9S dimer, and 11S dimer in external secretions which carries secretory component (S).
† IgA dimer and IgM contain J chain.
‡ ng = 10^{-9} g.

Table 3.3. Biological properties of major immunoglobulin classes in the human.

	IgG	IgA	IgM	IgD	IgE
Major characteristics	Most abundant Ig of internal body fluids particularly extravascular where it combats micro-organisms and their toxins	Major Ig in seromucous secretions where it defends external body surfaces	Very effective agglutinator; produced early in immune response – effective first-line defence against bacteremia	Mostly present on lymphocyte surface	Protection of external body surfaces. Recruits anti-microbial agents. Raised in parasitic infections. Responsible for symptoms of atopic allergy
Complement fixation					
Classical	++	–	+++	–	–
Alternative	–	+	–	–	–
Cross placenta	++	–	–	–	–
Sensitizes mast cells and basophils	–	–	–	–	+++
Binding to macrophages and polymorphs	+++	+	–	–	+

The diversity of Fcγ receptors

Since a wide variety of interactions between IgG complexes and different effector cells have been identified, we really should spend a little time looking at the membrane receptors for Fcγ which mediate these phenomena (figure 3.14). But first let us reiterate a point made earlier in figure 2.5 (p. 24). Simple occupancy of the receptor by its IgG ligand does *not* activate the cell; cellular activation is only triggered when the receptors are cross-linked by immune complexes containing more than one IgG molecule.



	FcγRI	FcγRIIA	FcγRIIB1	FcγRIIB2	FcγRIIC	FcγRIIIA	FcγRIIIB
Affinity (M^{-1})	hi 10^9	lo$<10^7$	lo$<10^7$	lo$<10^7$	lo$<10^7$	med 3×10^7	lo$<10^7$
Bind monomeric IgG	++	–	–	–	–	–	–
Bind aggregated IgG	+++	++	++	++	++	++	++
Present on:							
Monocytes	+	+	–	+	–	i	–
Macrophages	+	+	–	+	–	+	–
Dendritic cells	+	+	–	+	–	+	–
Neutrophils	i	+	–	+	–	+	+
Eosinophils	i	+	–	+	–	+	–
NK cells	–	–	–	–	+	+	–
B-cells	–	–	+	+	–	+	–
Mast cells	–	–	+	+	–	+	–

Figure 3.14: The structures and characteristics of leukocyte surface receptors for IgG Fc regions. bars = disulfide bridge; i = inducible. Although three separate genes and several different transcripts have been described for the high affinity FcγRI, only one of these, the FcγRIA transcript, has been convincingly demonstrated to produce an FcγRI protein. This possesses three immunoglobulin-like domains, and provides a higher affinity binding for IgG than the other types of FcγR which have only two immunoglobulin-like domains. Recent evidence suggests that FcγRI may also bind C-reactive protein. There are also three separate genes for the low affinity FcγRII. These all encode proteins and, due to splice variants, collectively produce a total of six expressed isoforms of FcγRII. They each possess two immunoglobulin-like domains but have diverse cytoplasmic domains which contain signaling motifs. The FcγRIIA isoform additionally exhibits genetic polymorphism, with either a histidine or arginine at position 131, which influences the affinity of the receptor for the different IgG subclasses. The His 131 variant is the only FcR on leukocytes which is capable of interacting with IgG2 and, perhaps because of this, polymorphisms of FcγRIIA appear to be risk factors for certain infectious and autoimmune diseases. The FcγRII family members have signaling motifs in their cytoplasmic tails which undergo phosphorylation by tyrosine kinases, either a stimulatory ITAM (*i*mmunoreceptor *t*yrosine-based *a*ctivation *m*otif; ●) in the case of FcγRIIA and FcγRIIC, or an inhibitory ITIM (*i*mmunoreceptor *t*yrosine-based *i*nhibitory *m*otif; ●) in the cytoplasmic tails of the FcγRIIB molecules. The other Fcγ receptors lack signaling motifs of their own and, in the case of FcγRI and FcγRIIIA, are therefore associated with the γ signaling homodimer which bears the necessary ITAMs. FcγRIIIA in NK cells can alternatively associate with the ITAM-containing ζ chain signaling molecule also used by the T-cell receptor (TCR). Homologous ITAMs are present in the cytoplasmic regions of the Ig α/β heterodimer (see figure 4.3), the CD3 γ, δ, ε molecules and in the β chain of one of the versions of the high affinity IgE receptor. It is not known how FcγRIIIB transmits signals into the cell, but this GPI anchored molecule may associate with so-called cell membrane 'rafts' which contain signaling molecules. Genetically inherited polymorphisms of both FcγRIIIA and FcγRIIIB may be associated with the autoimmune disease systemic lupus erythematosus (SLE) (cf. p. 425).

Specific Fc receptors may exist for all the classes of antibody, although receptors for IgD and IgM are not particularly well characterized. For IgG a quite extensive family of Fc receptors has emerged, the nine known genes encoding four major groups of receptors FcγRI, FcγRII, FcγRIII and FcγRn (or, more simply, FcRn). Within these groups there are a number of splice variants and inherited polymorphic forms. All the Fcγ receptors display immunoglobulin-like domains (p. 45) on the extracellular surface and the leukocyte Fcγ receptors (FcγRI, FcγRII, FcγRIII) utilize these for binding Ig Fc regions, just as other molecules of related structure (the immunoglobulin 'superfamily', p. 245) are mutually attracted through domain interactions.

FcγRI (CD64) is constitutively present on monocytes, macrophages and dendritic cells, and is induced on neutrophils following their activation by IFNγ and G-CSF (*g*ranulocyte *c*olony-*s*timulating *f*actor). Conversely, FcγRI can be downregulated in response to IL-

4 and IL-13. Structurally, it consists of an IgG-binding α chain which lacks cytoplasmic signaling domains and a γ chain homodimer containing *i*mmunoreceptor *t*yrosine-based *a*ctivation *m*otifs (ITAMs) used to transmit an activation signal into the cell. It binds monomeric IgG avidly to the surface of the cell thus sensitizing it for subsequent encounter with antigen. Its main roles are probably in facilitating phagocytosis, antigen presentation and in mediating extracellular killing of target cells coated with IgG antibody, a process referred to as *a*ntibody-*d*ependent *c*ellular *c*ytotoxicity (ADCC; p. 332). It might also be concerned with the overall regulation of IgG levels in the body, since the *catabolic rate* appears to depend directly upon the total IgG concentration and one might speculate that endocytosis of IgG attached to FcγRI, the only leukocyte Fc receptor to have a high affinity for monomeric IgG, could contribute significantly to this degradation. On the other hand, *synthesis* is largely governed by antigen stimulation, so that in germ-free animals, for example, IgG levels are extremely low but rise rapidly on transfer to a normal environment.

Unlike the single isoform of FcγRI, there are six expressed isoforms of **FcγRII (CD32)** which collectively are present on the surface of most types of leukocyte (figure 3.14). Binding of monomeric IgG to these receptors is insignificant. This lends itself to a cheeky play of outstanding simplicity for the uptake of immune complexes: because of the geometric increase in binding strength of polymeric vs monomeric ligands (cf. the 'bonus effect of multivalency', p. 87), complexes bind really well to these receptors and are selectively adsorbed to the cell surface in the face of competition from the dauntingly high concentrations of monomeric IgG in the body fluids. Furthermore, because of their fixed spatial relationship within the immune complex, the bound IgG molecules bring about the cross-linking of Fc receptors mandatory for cell signaling. Thus the binding of IgG complexes triggers phagocytic cells and may provoke thrombosis through their reaction with platelets. The FcγRIIA mediates phagocytosis and ADCC whilst the FcγRIIB2 (and FcγRIII) efficiently mediates endocytosis leading to antigen presentation. FcγRIIB1 on B-cells does not endocytose immune complexes and therefore B-cells principally present only their cognate antigen following ligation and endocytosis of the *B*-cell *r*eceptor (BCR). In fact, the FcγRIIB molecules have cytoplasmic domains which contain *i*mmunoreceptor *t*yrosine-based *i*nhibitory *m*otifs (ITIMs) and their occupation leads to *downregulation* of cellular responsiveness. In the case of the B-cell this mediates the negative-feedback effect of IgG on antibody production (cf. p. 201). Thus, whereas the isoforms on phagocytic cells are associated with ligand internalization, that on the B-cell fails to internalize but concentrates instead on lymphocyte regulation.

The two ***FcγRIII (CD16)*** genes encode the isoforms FcγRIIIA and FcγRIIIB which have a medium and low affinity for IgG, respectively. FcγRIIIA is found on most types of leukocyte, whereas FcγRIIIB is restricted mainly to neutrophils and is unique amongst the Fc receptors in being attached to the cell membrane by a glycosylphosphatidylinositol (GPI) anchor rather than a transmembrane segment. FcγRIIIA is known to be associated with the γ chain signaling dimer on monocytes and macrophages, and with either ζ and/or γ chain signaling molecules in NK cells, and its expression is upregulated by transforming growth factor β (TGFβ) and downregulated by IL-4. With respect to their functions, FcγRIIIA is largely responsible for mediating ADCC by NK cells and the clearance of immune complexes from the circulation by macrophages. For example, the clearance of IgG-coated erythrocytes from the blood of chimpanzees was essentially inhibited by the monovalent Fab fragment of a monoclonal anti-FcγIII (work out why the Fab fragment was used). FcγRIIIB cross-linking stimulates the production of superoxide by neutrophils.

The fourth major type of FcγR is not present on leukocytes but instead is found on epithelial cells. It is referred to as **FcRn** due to its original description as a **n**eonatal receptor, although it is now known to be present also in the adult. This molecule has a *m*ajor *h*istocompatibility *c*omplex (MHC)-like structure and its α chain associates with β_2-microglobulin. It acts as an intracellular trafficking receptor and at least four major activities have been proposed for this molecule. It is fairly well established that it transfers maternal circulating IgG across the placenta to the fetus (figure 3.15a) and, at least in rodents, colostral and maternal milk IgG from the intestine to the circulation of the neonate (figure 3.15b). IgG is alone amongst the Ig classes in its ability to cross the human placenta, thereby providing a major life-line of defense not only for the fetus *in utero* but also for the first few weeks of the newborn baby's life. Transfer of IgG across the gut mucosa in the neonate may further reinforce this protection. FcRn may additionally be involved in maintaining steady state levels of circulating IgG and in the bidirectional transport of IgG across the adult intestinal epithelium and possibly across mucosal surfaces at other sites in the body. Binding of IgG to FcRn is regulated by pH,

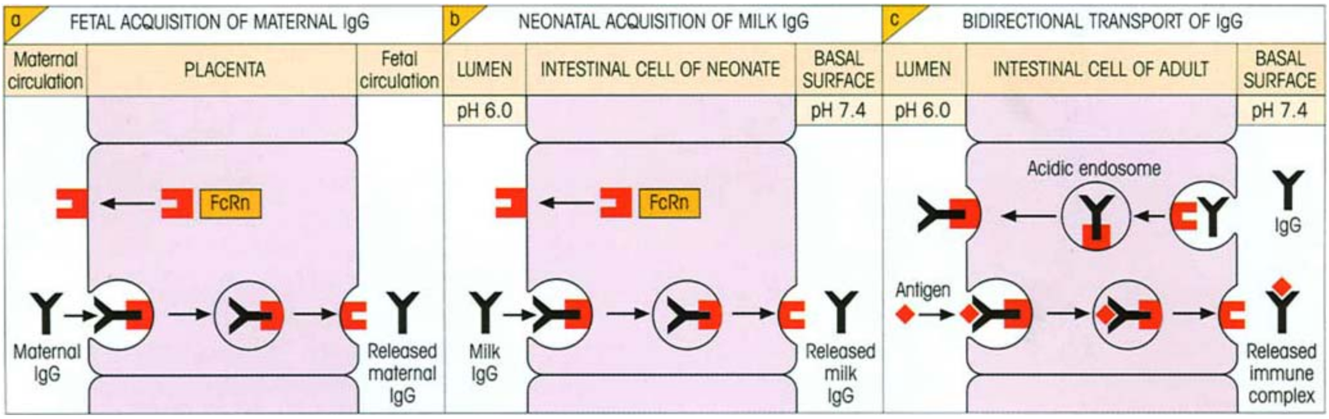


Figure 3.15. The epithelial cell surface receptor for IgG Fc regions. (a) The FcRn receptor is present in the placenta where it fulfils the important task of transferring maternal IgG to the fetal circulation. This will provide important protection prior to the generation of immunocompetence in the fetus. Furthermore, it is self-evident that any infectious agent which might reach the fetus *in utero* will have had to have passed through the mother first, and the fetus will rely upon the mother's immune system to have produced IgG with appropriate binding specificities. This maternal IgG also provides protection for the neonate, because it takes some weeks following birth before the transferred IgG is eventually all catabolized. (b) It has been clearly demonstrated in rodents, although remains speculative in humans, that there is epithelial transport of IgG from maternal milk across the intestinal cells of the newborn. IgG binds to FcRn at pH 6.0, is taken into the cell within a clathrin-coated vesicle and released at the pH of the basal surface. The directional movement of IgG is achieved by the asymmetric pH effect on Ig–receptor interaction. FcRn has homology to MHC class I molecules and contains β_2-microglobulin (cf. p. 70). Knockout mice lacking FcRn are incapable of acquiring maternal Ig as neonates. Furthermore, they have a grossly shortened IgG half-life, suggesting that FcRn may also serve a role as a protective receptor which prevents degradation of IgG and then recycles it to the circulation. The IgG half-life is unusually long compared with that of IgA and IgM and this enables the response to antigen to be sustained for many months following infection. (c) An additional, and still to be proven, role of FcRn may be as a bidirectional shuttle receptor. IgG binding on the nonluminal side of the epithelial cell may occur, following endocytosis, within the more favorable pH of acidic endosomes. This receptor may thus provide a mechanism for mucosal immunosurveillance, traveling back and forth across the epithelial cell, delivering IgG to the intestinal lumen and then returning the same antibodies in the form of immune complexes for the stimulation of B-cells by follicular dendritic cells.

with high affinity binding at acidic pH but very little at neutral or alkali pH. Unlike the transcytosis across epithelial cells mediated by the pIgA-R (see below), FcRn does not undergo proteolytic cleavage and therefore has the potential for generating bidirectional traffic (figure 3.15c).

Nonprecipitating 'univalent' antibodies

IgG has two combining sites for antigen. However, 5–15% of the IgG in all antiserums appears to consist of nonprecipitating asymmetric molecules with a single effective binding site. The other site is blocked stereochemically by a mannose-rich carbohydrate in the Fab region.

Immunoglobulin A guards the mucosal surfaces

IgA appears selectively in the seromucous secretions, such as saliva, tears, nasal fluids, sweat, colostrum, and secretions of the lung, genitourinary and gastrointestinal tracts, where it clearly has the job of defending the exposed external surfaces of the body against attack by microorganisms. This responsibility is clearly taken seriously since approximately 40 mg of secretory IgA/kg body weight is transported daily through the human intestinal crypt epithelium to the mucosal surface as compared with a *total* daily production of IgG of 30 mg/kg.

The IgA is synthesized locally by plasma cells and dimerized intracellularly together with a cysteine-rich polypeptide called J chain, of molecular weight 15 000. The dimeric molecule is effectively tetravalent and will have a much higher binding avidity for polymeric antigens than the monomeric form due to the bonus effect of multivalency (cf. p. 87). If dimerization occurred randomly *after* secretion, dimers of mixed specificity would be formed which would be no more effective in combining with antigen than the monomers. The dimeric IgA binds strongly through its J chain to a receptor for polymeric Ig (**poly-Ig receptor, pIgR**, which also binds polymeric IgM) present in the membrane of mucosal epithelial cells. The complex is then actively endocytosed, transported across the cytoplasm and secreted into the external body fluids after cleavage of the pIgR peptide chain. The fragment

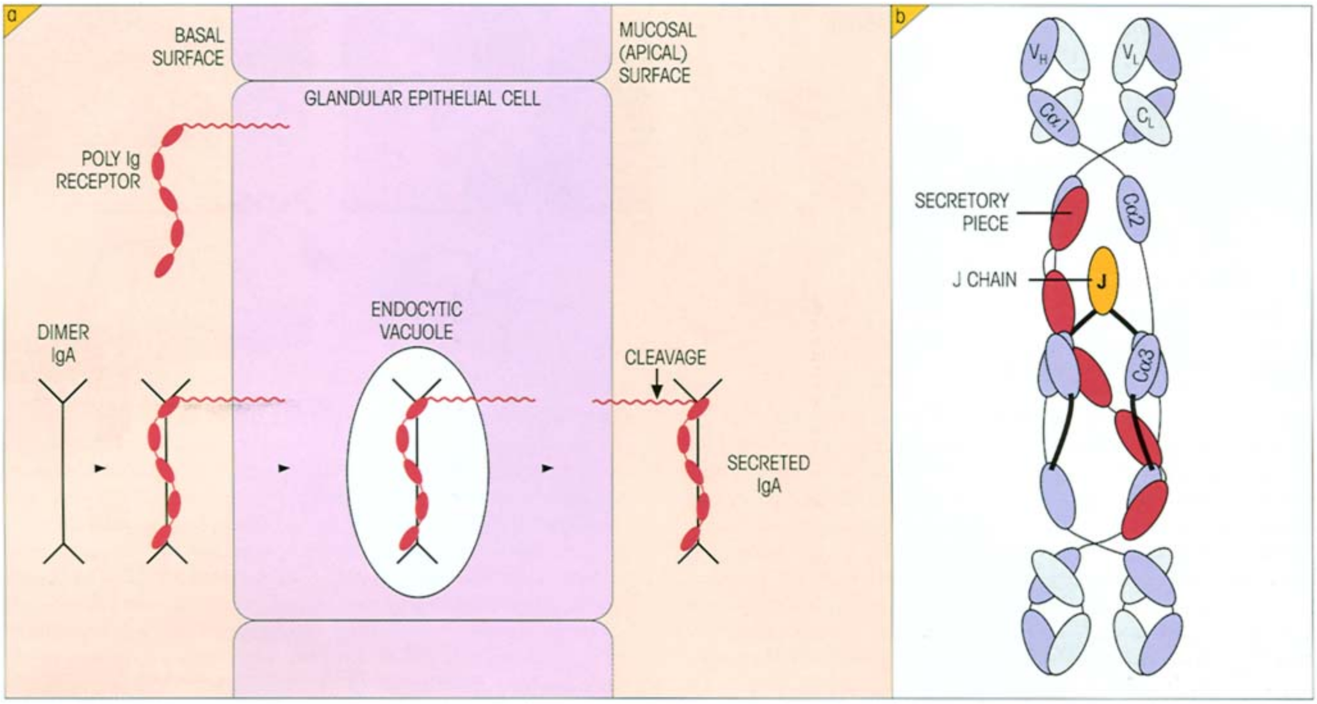


Figure 3.16. Secretory IgA. (a) The mechanism of IgA secretion at the mucosal surface. The mucosal cell synthesizes a receptor for polymeric Ig (pIgR) which is inserted into the basal membrane. Dimeric IgA binds to this receptor and is transported via an endocytic vacuole to the apical surface. Cleavage of the receptor releases secretory IgA still attached to part of the receptor termed the secretory piece. Note how the receptor cleavage introduces an asymmetry which drives the transport of IgA dimers to the mucosal surface (in quite the opposite direction to the transcytosis of milk IgG in figure 3.15). (b) Schematic view of the structure of secreted IgA. The J chain, which is an integral part of secreted polymeric Ig (IgA and IgM), forms disulfide bonds with the penultimate residue of the Cα3 domain which is a cysteine (Cys 495). Covalent linkage of the J chain seems to be critical for the initial binding to the polymeric Ig receptor.

of the receptor remaining bound to the IgA is termed secretory component and the whole molecule, **secretory IgA** (figure 3.16). The reader is strongly recommended to turn to figure 8.9 (p. 156) for a dramatic demonstration of secretory IgA held in the surface mucus of intestinal mucosal epithelial cells.

The function of the secretory piece may be to protect the IgA hinge from bacterial proteases. It would also be nice to think that it acted as a molecular Teflon to endow the IgA dimer with 'nonstick' potential, since IgA antibodies function by inhibiting the adherence of coated microorganisms to the surface of mucosal cells, thereby preventing entry into the body tissues. They will also combine with the myriad soluble antigens of dietary and microbial origin to block their access to the body. Aggregated IgA binds to polymorphs and can also activate the alternative (figure 2.3), as distinct from the classical, complement pathway, largely through its abundant carbohydrate groups. This may account for reports of a synergism between IgA, complement and lysozyme in the killing of certain coliform organisms, where it is supposed that complement-induced disruption of the outer surface permits access of the enzyme to the peptidoglycan wall.

Plasma IgA is predominantly monomeric and, since this form is a relatively poor activator of complement, it seems likely that the body uses it for the direct neutralization of any antigens which breach the epithelial barrier to enter the circulation in appreciable quantities. However, it has additional functions which are mediated through the FcαR (CD89) for IgA present on monocytes, macrophages, neutrophils and subpopulations of both T- and B-lymphocytes. Structurally, this receptor most closely resembles FcγRIIIA in that it has two immunoglobulin domains and associates with a γ chain signaling homodimer. Following cross-linking, the receptor can activate endocytosis, phagocytosis, inflammatory mediator release and ADCC. Expression of the FcαR on monocytes is strongly upregulated by bacterial lipopolysaccharide.

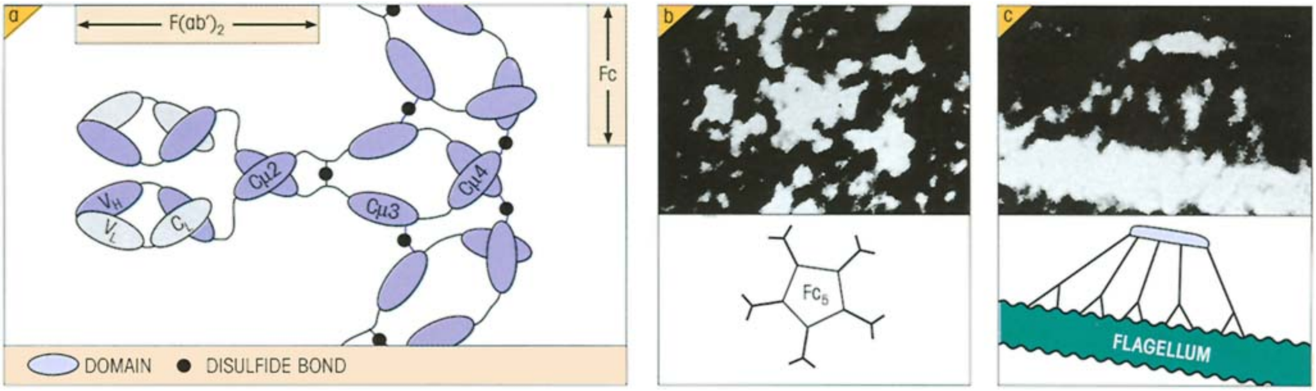


Figure 3.17. The structure of IgM. (a) The arrangement of domains in one of the five subunits showing how the pentamer is built up through the disulfide linkages between the Cys 414 in the Cμ3 domains and between the Cys 575, the penultimate C-terminal residue in the 19-amino acid tailpiece which follows Cμ4 (after Hilschman & Feinstein). Without too much aggravation, we hope the reader will appreciate that the hinge region in IgG (cf. figure 3.13) is replaced by a rigid pair of extra domains (Cμ2), while Cμ3 and Cμ4 domains in IgM are structurally equivalent to the Cγ2 and Cγ3 regions, respectively, in IgG. (b) The structure as shown by electron microscopy of a human Waldenström's macroglobulin in free solution adopting a 'star'-shaped configuration. (c) The structure as revealed in an electron microscope preparation of specific sheep IgM antibody bound to *Salmonella paratyphi* flagellum where the immunoglobulin has assumed a 'crab-like' conformation in establishing its links with antigen. With the F(ab')$_2$ arms bent out of the plane of the central Fc$_5$ region, the Cμ3 complement binding domains are now readily accessible to the first component of complement (cf. p. 22). The Fc$_5$ constellation obtained by papain cleavage can activate complement directly. (Electron micrographs are negatively stained preparations of magnification ×2 000 000, i.e. 1 mm represents 0.5 nm; kindly provided by Dr A. Feinstein and Dr E.A. Munn.)

Immunoglobulin M provides a defense against bacteremia

In the past referred to as the macroglobulin antibodies because of their high molecular weight, IgM molecules are polymers of five four-peptide subunits each bearing an extra C_H domain. As with IgA, a single J chain is incorporated into the pentamer which has the structure shown in figure 3.17a. Under negative staining in the electron microscope, the free molecule in solution assumes a 'star' shape but, when combined as an antibody with an antigenic surface membrane, it can adopt a 'crab-like' configuration (figure 3.17b and c) in which multiple Fc regions are now accessible to C1q which is bound most firmly. A small proportion of circulating IgM exists in a hexameric form which lacks J chain but is up to 20 times more effective in activating complement-mediated lysis than the pentamer. Circulating monomeric IgM (i.e. a single four-peptide unit) is present during chronic infections but is unable to activate complement. The theoretical combining valency of the pentamer is of course 10 but this is only observed on interaction with small haptens; with larger antigens the effective valency falls to 5 and this must be attributed to some form of steric restriction due to lack of flexibility in the molecule. IgM antibodies tend to be of relatively low affinity as measured against single determinants (haptens) but, because of their high valency, they bind with quite respectable avidity to antigens with multiple epitopes (bonus effect of multivalency, p. 87). For the same reason, these antibodies are extremely efficient agglutinating and cytolytic agents and, since they appear early in the response to infection and are largely confined to the bloodstream, it is likely that they play a role of particular importance in cases of bacteremia. The isohemagglutinins (anti-A, anti-B) and many of the 'natural' antibodies to microorganisms are usually IgM; antibodies to the typhoid 'O' antigen (endotoxin) and the 'WR' antibodies in syphilis are also found in this class. IgM would appear to precede IgG in the phylogeny of the immune response in vertebrates.

Monomeric IgM, with a hydrophobic sequence stitched into the C-terminal end of the heavy chain to anchor the molecule in the cell membrane, is the major antibody receptor used by B-lymphocytes to recognize antigen (cf. figure 2.10).

Immunoglobulin D is a cell surface receptor

This class was recognized through the discovery of a myeloma protein which did not have the antigenic specificity of IgG, A or M, although it reacted with antibodies to immunoglobulin light chains and had the

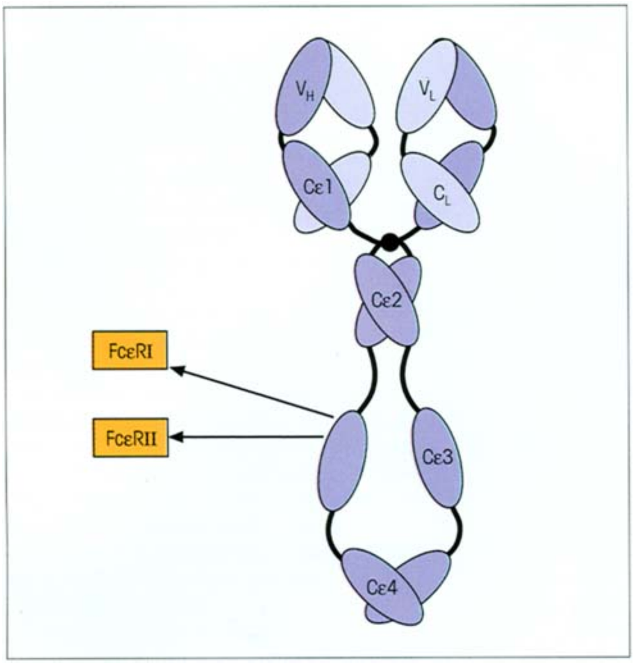


Figure 3.18. Domain structure of IgE. Note the general similarity in structure to the IgM basic unit in figure 3.17 with the IgG hinge replaced by the extra Cε2 paired domains, but also the lack of association of the penultimate C-terminal domains (in this case Cε3) which is a consistent feature of all Ig classes. The α chain of the high affinity mast cell receptor for IgE (FcεRI) binds very avidly to a single site located in positions 301–304 of the Cε3 domain. The low affinity FcεRII on inflammatory cells and B-lymphocytes requires the presence of both Cε3 domains for significant binding to IgE. 'Knockout' mice lacking the FcεRI α chain do not express the receptor and cannot produce anaphylactic reactions; conclusion, the low affinity FcεRII receptor which is expressed does not contribute significantly to IgE/mast cell-mediated anaphylaxis.

basic four-peptide structure. The hinge region is particularly extended and, although protected to some degree by carbohydrate, it may be this feature which makes IgD, among the different immunoglobulin classes, uniquely susceptible to proteolytic degradation, and account for its short half-life in plasma (2.8 days). Nearly all the IgD is present, together with IgM, on the surface of a proportion of B-lymphocytes where it seems likely that they may operate as mutually interacting antigen receptors for the control of lymphocyte activation and suppression.

Immunoglobulin E triggers inflammatory reactions

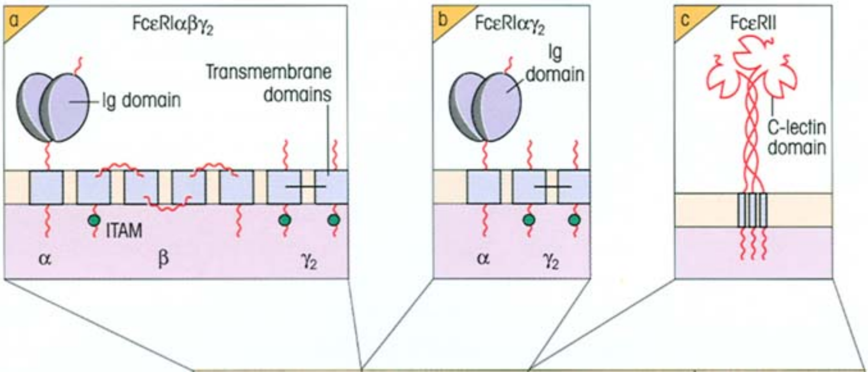
Only very low concentrations of IgE are present in serum and only a very small proportion of the plasma cells in the body are synthesizing this immunoglobulin. It is not surprising, therefore, that so far only a handful of IgE myelomas have been recognized compared with tens of thousands of IgG paraproteinemias. IgE antibodies (figure 3.18) remain firmly fixed for an extended period when injected into human skin, bound to the high affinity FcεRI receptor on mast cells (figure 3.19). Contact with antigen leads to degranulation of the mast cells with release of preformed vasoactive amines and cytokines, and the synthesis of a variety of inflammatory mediators derived from arachidonic acid (cf. figure 1.15). This process is responsible for the symptoms of hay fever and of extrinsic asthma when patients with atopic allergy come into contact with the allergen, e.g. grass pollen.

The main *physiological* role of IgE would appear to be protection of anatomic sites susceptible to trauma and pathogen entry by local recruitment of plasma factors and effector cells through the **triggering of an acute inflammatory reaction**. Infectious agents penetrating the IgA defenses would combine with specific IgE on the mast cell surface and trigger the release of vasoactive agents and factors chemotactic for granulocytes, so leading to an influx of plasma IgG, complement, neutrophils and eosinophils (cf. p. 272). In such a context, the ability of eosinophils to damage IgG-coated helminths and the generous IgE response to such parasites would constitute an effective defense.

The low affinity FcεRII receptor, CD23, is present on many different types of hematopoietic cells (figure 3.19c). Its primary function appears to be in the regulation of IgE synthesis by B-cells, with a stimulatory role at low concentrations of IgE and an inhibitory role at high concentrations. It can also facilitate phagocytosis of IgE opsonized antigens.

Immunoglobulins are further subdivided into subclasses

Antigenic analysis of IgG myelomas revealed further variation and showed that they could be grouped into four isotypic **subclasses** termed IgG1, IgG2, IgG3 and IgG4. The differences all lie in the heavy chains which have been labeled γ1, γ2, γ3 and γ4, respectively. These heavy chains show considerable homology and have certain structures in common with each other—the ones which react with specific anti-IgG antisera—but each has one or more additional structures characteristic of its own subclass arising from differences in primary amino acid composition and in interchain disulfide bridging. These give rise to differences in biological behavior which are summarized in table 3.4.



	FcεRIαβγ2	FcεRIαγ2	FcεRIIA (CD23)	FcεRIIB (CD23)
Affinity (M^{-1})	v.hi 10^{10}	v.hi 10^{10}	lo<10^{7}	lo<10^{7}
Bind monomeric IgG	–	–	–	–
Bind aggregated IgG	–	–	–	–
Bind monomeric IgE	++	++	+	+
Present on:				
Monocytes	–	+	–	+
Macrophages	–	+	–	+
Dendritic cells	–	+	–	+
Neutrophils	–	–	–	+
Eosinophils	–	+	+	+
NK cells	–	–	–	?
T-cells	–	–	–	+
B-cells	–	–	+	+
Mast cells	++	++	–	?

Figure 3.19. The structures and characteristics of surface receptors for IgE Fc regions. Bars signify disulfide bridges. The high affinity receptor for IgE, FcεRI, is found only on basophils and mast cells (and some non-T-, non-B-lymphocytes) in the mouse. However, humans also express a version of FcεRI that lacks an associated four-transmembrane region β chain, and this receptor is expressed, albeit at lower levels than on mast cells, on a number of different cell types. More recently acquired data on FcεRI structure are presented in Figure 16.1. The high affinity binding site for IgE on these receptors involves a hydrophobic patch of four tryptophans at the top of the molecule. The low affinity receptor for IgE, FcεRII (CD23), differs from most other types of Fc receptors in that it utilizes C-type lectin domains rather than Ig-type domains. It is present on the cell surface as a trimer, but the details of how it transduces signals into the cell are incompletely understood. There are two versions of FcεRII which differ by seven amino acids in their cytoplasmic tails. Whilst FcεRIIA is essentially restricted to B-cells and eosinophils, FcεRIIB shows a much broader distribution.

	IgG1	IgG2	IgG3	IgG4
Serum concentration (mg/ml)	9	3	1	0.5
% Total IgG in normal serum	67	22	7	4
Serum half-life (days)	23	23	8	23
Complement activation (classical pathway)*	++	+	+++	±
Binding to monocyte/macrophage Fc receptors	+++	±	+++	+
Ability to cross placenta	+++	±	+++	+++
Spontaneous aggregation	–	–	+++	
Binding to staphyloccal protein A	+++	+++	±	+++
Binding to staphyloccal protein G	+++	+++	+++	+++
Gm allotypes	a,z,f,x	n	b0,b1,b3, g,s,t, etc.	4a,4b

Table 3.4. Comparison of human IgG subclasses. *The very poor complement-fixing ability of IgG4 cannot be ascribed to its rigid hinge region since substitution of serine 331 with proline (as in IgG1 and IgG3) endows the molecule with excellent C1q-binding and complement-mediated lytic capability. Intriguingly, substitution of proline 331 with serine in IgG1 maintains C1q-binding ability but grossly diminishes lytic activity, a puzzle still to be resolved.

Two subclasses of IgA have also been found, of which IgA1 constitutes 80–90% of the total. The IgA2 subclass exists in two allotypic forms, one of which (IgA2m(1)) is unusual in that it lacks interchain disulfide bonds between heavy and light chains. Class and subclass variation is not restricted to human immunoglobulins but is a feature of all the mammals so far studied: monkey, sheep, rabbit, guinea-pig, rat and mouse.

SUMMARY

The basic structure is a four-peptide unit

• Immunoglobulins (Ig) have a basic four-peptide structure of two identical heavy and two identical light chains joined by interchain disulfide links.

• Papain splits the molecule at the exposed flexible hinge region to give two identical univalent antigen-binding fragments (Fab) and a further fragment (Fc). Pepsin proteolysis gives a divalent antigen-binding fragment $F(ab')_2$ lacking the Fc.

Amino acid sequences reveal variations in immunoglobulin structure

• There are perhaps 10^8 or more different Ig molecules in normal serum.

• Analysis of myeloma proteins, which are homogeneous Ig produced by single clones of malignant plasma cells, has shown the N-terminal region of heavy and light chains to have a variable amino acid structure and the remainder to be relatively constant in structure.

Immunoglobulin genes

• Clusters of genes on three different chromosomes encode κ, λ and heavy Ig chains, respectively. In each cluster in humans there are approximately 30–50 functional variable region (*V*) genes and around four to six small *J* minisegments. Heavy chain clusters in addition contain of the order of 25 *D* minigenes. There is a single gene encoding each of the nine different class and subclass constant regions.

• A special splicing mechanism involving mutual recognition of 5′ and 3′ flanking sequences, catalysed by recombinase enzymes, effects the *DJ*, *VD* and *VJ* translocations.

Structural variants of the basic Ig molecule

• Isotypes are Ig variants based on different heavy chain constant structures, all of which are present in each individual; examples are the Ig classes IgG, IgA, etc.

• Allotypes are heavy chain variants encoded by allelic (alternative) genes at single loci and are therefore genetically distributed; examples are Gm groups.

• An idiotype is the collection of antigenic determinants on an antibody, usually associated with the hypervariable regions, recognized by other antigen-specific receptors, either antibody (the anti-idiotype) or T-cell receptors.

• The variable region domains bind antigen, and three hypervariable loops (termed complementarity determining regions) on the heavy chain and three on the light chain form the antigen-binding site.

• The constant region domains of the heavy chain (particularly the Fc) carry out a secondary biological function after the binding of antigen, e.g. complement fixation and macrophage binding.

Immunoglobulin classes and subclasses

• In the human there are five major types of heavy chain giving five classes of Ig. IgG is the most abundant Ig in the extravascular fluids where it neutralizes toxins and combats microorganisms by fixing complement via the C1 pathway and facilitating the binding to phagocytic cells by receptors for C3b and Fcγ. It crosses the placenta in late pregnancy and the intestine in the neonate.

• Various Fcγ receptors are specialized for different functions such as phagocytosis, antibody-dependent cellular cytotoxicity, placental transport and B-lymphocyte regulation.

• IgA exists mainly as a monomer (basic four-polypeptide unit) in plasma, but in the seromucous secretions, where it is the major Ig concerned in the defense of the external body surfaces, it is present as a dimer linked to a secretory component.

• IgM is most commonly a pentameric molecule although a minor fraction is hexameric. It is essentially intravascular and is produced early in the immune response. Because of its high valency it is a very effective bacterial agglutinator and mediator of complement-dependent cytolysis and is therefore a powerful first-line defense against bacteremia.

• IgD is largely present on the lymphocyte and functions together with IgM as the antigen receptor on naive B-cells.

• IgE binds firmly to mast cells and contact with antigen leads to local recruitment of antimicrobial agents through degranulation of the mast cells and release of inflammatory mediators. IgE is of importance in certain parasitic infections and is responsible for the symptoms of atopic allergy.

• Further diversity of function is possible through the subdivision of classes into subclasses based on structural differences in heavy chains all present in each normal individual.

See the accompanying website (**www.roitt.com**) for multiple choice questions.